DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

STATISTICS FOR DATA ANALYTICS

Unit-wise Question Bank with Answers

Course Title	Statistics for Data Analytics	**Course Code**	IM241AT
Semester	IV	**Units Covered**	I – V
Questions	Part A: 101 Part B: 85	**Scheme**	2022

Course Outcomes

CO1	Describe and report a data set using data analysis, presentation and interpretation techniques to understand various phenomena in science and engineering.
CO2	Apply various statistical processing techniques to handle a set of data to estimate probabilities.
CO3	Apply an appropriate statistical tool and analyse a specific set of data to estimate and draw conclusions about population parameters.
CO4	Draw inferences about population parameters and relations between variables based on analysis of sample data.

Notes:

1. The questions are arranged unit by unit. Within each unit, Part A has short-answer questions (1–2 marks) and Part B has descriptive and numerical questions.
2. Every question is followed by its answer. Numerical answers show the working, and the final result is stated after **Answer:**.
3. Quartiles are found by the $(n+1)p$ position rule with linear interpolation (the rule used by Montgomery and by Minitab). Where a short question is usually answered by the median-of-halves rule, that value is also given.
4. Values of z, t and χ^2 are computed to four decimals; they agree with the statistical tables to the precision the tables print.
5. M: Marks; BT: Bloom's Taxonomy level; CO: Course Outcome.

UNIT I – DATA SUMMARY, PRESENTATION AND CONCEPTS OF PROBABILITY

Data types; stem-and-leaf diagrams, histograms, box plots, radar charts; sample spaces and events; addition, conditional, multiplication and total probability rules; independence; Bayes' theorem

Sl. No.	Questions	M	BT	CO
	PART – A (Short-answer questions)			
1	The 40th percentile in a given set of 140 data values occurs at which position?	1	L2	CO1

Position of the pth percentile $= \dfrac{p}{100} \times n = \dfrac{40}{100} \times 140 = 56.$

Answer: the 56th position.

2	The lengths of 91 items are grouped as below. In which class does the median lie?	1	L2	CO1

Class	21–30	31–40	41–50	51–60	61–70
Frequency	12	16	22	25	18

Cumulative frequencies: 12, 28, 50, 75, 93. The frequencies add up to $N = 93$ (the stem says 91, but the table totals 93; the answer is the same either way). The median is the $N/2 = 46.5^{\text{th}}$ observation. Since $28 < 46.5 \leq 50$, it lies in the third class.

Answer: the median class is 41–50.

3	For the same data on the length of 93 wooden planks, in which class does the mode lie?	1	L2	CO1

The modal class is the class with the highest frequency. The largest frequency is 25.

Answer: the modal class is 51–60.

4	For the data set 62, 65, 68, 70, 72, 74, 76, 78, 80, 82, 96, 101, identify the median and the first quartile.	2	L2	CO1

$n = 12$. Median $= \dfrac{74 + 76}{2} = 75$ (average of the 6th and 7th values). Q_1 is at position $(n+1)/4 = 3.25$: $Q_1 = 68 + 0.25(70 - 68) = 68.5$. (By the median-of-halves rule, $Q_1 = \frac{68+70}{2} = 69$.)

Answer: Median $= 75$, $Q_1 = 68.5$ (69 by halves).

5	For the data set 62, 65, 68, 70, 72, 74, 76, 78, 80, 82, 96, 101, identify the third quartile and the interquartile range (IQR).	2	L2	CO1

Q_3 is at position $3(n+1)/4 = 9.75$: $Q_3 = 80 + 0.75(82 - 80) = 81.5$, so IQR $= Q_3 - Q_1 = 81.5 - 68.5 = 13$. (By halves: $Q_3 = \frac{80+82}{2} = 81$ and IQR $= 81 - 69 = 12$.)

Answer: $Q_3 = 81.5$, IQR $= 13$ (81 and 12 by halves).

6	For the ordered data set 38, 39, 41, 45, 46, 47, 48, 48, 51, what is the interquartile range?	2	L2	CO1

$n = 9$. Q_1 at position 2.5: $Q_1 = 39 + 0.5(41 - 39) = 40$; Q_3 at position 7.5: $Q_3 = 48 + 0.5(48 - 48) = 48$. IQR $= 48 - 40 = 8$. (By halves, $Q_1 = \frac{39+41}{2} = 40$, $Q_3 = \frac{48+48}{2} = 48$, IQR $= 8$.)

Answer: IQR $= 8$.

7	Find the median of the ungrouped data 86, 52, 49, 43, 35, 31, 30, 11.	2	L1	CO1

Sorted: 11, 30, 31, 35, 43, 49, 52, 86. With $n = 8$ the median is the average of the 4th and 5th values: $\dfrac{35 + 43}{2} = 39$.

Answer: 39.

8	What inference can be drawn from a data set with different values of mean, median and mode?	2	L2	CO1

When the mean, median and mode differ, the distribution is not symmetric (it is skewed). If mode $<$ median $<$ mean, the data are right (positively) skewed with a long upper tail; if mean $<$ median $<$ mode, they are left (negatively) skewed. The mean is pulled toward the tail, so the median is the better measure of the centre.

9	In what way is a stem-and-leaf diagram better than a histogram?	2	L2	CO1

A stem-and-leaf diagram shows the shape of the distribution like a histogram, but it also keeps every original data value. So the median, quartiles, mode and individual values can be read directly from it; a histogram shows only class frequencies.

10	For the data set 45, 41, 48, 39, 52, 38, 46, 51, 47, 48, what is the interquartile range?	2	L2	CO1

Sorted: 38, 39, 41, 45, 46, 47, 48, 48, 51, 52 ($n = 10$). Q_1 at position 2.75: $39 + 0.75(41 - 39) = 40.5$; Q_3 at position 8.25: $48 + 0.25(51 - 48) = 48.75$. IQR $= 8.25$. (By halves, $Q_1 = 41$, $Q_3 = 48$ and IQR $= 7$.)

Answer: IQR $= 8.25$ (7 by halves).

11	The weights (in pounds) of 30 newborn babies are listed below. Find Q_1. 5.5, 5.7, 5.8, 6.0, 6.1, 6.1, 6.3, 6.4, 6.5, 6.6, 6.7, 6.7, 6.7, 6.9, 7.0, 7.0, 7.0, 7.1, 7.2, 7.2, 7.4, 7.5, 7.7, 7.7, 7.8, 8.0, 8.1, 8.1, 8.3, 8.7	2	L3	CO1

Position of $Q_1 = (n+1)/4 = 31/4 = 7.75$. The 7th value is 6.3 and the 8th is 6.4, so $Q_1 = 6.3 + 0.75(6.4 - 6.3) = 6.375$ lb.

Answer: $Q_1 = 6.375$ lb.

12	A lot of 100 semiconductor chips contains 25 defective ones. Two are selected at random without replacement. What is the probability that both are defective?	2	L3	CO2

$$P(\text{both defective}) = \frac{25}{100} \times \frac{24}{99} = \frac{600}{9900} = \frac{2}{33} = 0.0606.$$

Answer: $2/33 \approx 0.0606$.

13	X is the number shown when a fair six-faced die is thrown and Y is the number of heads when a coin is tossed once. State the possible values of X and Y.	2	L1	CO2

$X \in \{1, 2, 3, 4, 5, 6\}$ and $Y \in \{0, 1\}$.

14	Let $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 6, 8\}$, $C = \{6, 7, 8, 9\}$, $D = \{4, 5, 6, 7\}$ and $S = A \cup B \cup C \cup D$, with all elements of S equally likely. Find $P(A' \cap B)$.	2	L3	CO2

$S = \{1, 2, \ldots, 9\}$, so $|S| = 9$. $A' = \{5, 6, 7, 8, 9\}$ and $A' \cap B = \{6, 8\}$.

Answer: $P(A' \cap B) = 2/9 \approx 0.2222$.

15	What is the sample space for the number of calls repeatedly placed to a busy phone line until a connection is achieved?	1	L1	CO2

$S = \{1, 2, 3, \ldots\}$, a countably infinite (discrete) sample space.

16	A part is equally likely to have been produced on any one of six cutting tools. Find the probability that the part is not from tool 4.	1	L2	CO2

$$P(\text{not tool 4}) = 1 - \frac{1}{6} = \frac{5}{6} \approx 0.8333.$$

17	Cooking oil is produced in two varieties, mono- and polyunsaturated, from two sources, canola and corn. The numbers of bottles at a supermarket are: **Canola** / **Corn** Mono: 7 / 13 Poly: 93 / 77 If a bottle is selected at random, what is the probability that it is polyunsaturated?	2	L3	CO2

Total bottles $= 7 + 13 + 93 + 77 = 190$; polyunsaturated $= 93 + 77 = 170$.

$$P(\text{poly}) = \frac{170}{190} = \frac{17}{19} = 0.8947.$$

18	With A, B, C, D and S as defined in Part A Q14, find $P(A' \mid D)$.	2	L3	CO2

$D = \{4, 5, 6, 7\}$ and $A' \cap D = \{5, 6, 7\}$. Since all outcomes are equally likely, $P(A' \mid D) = \dfrac{|A' \cap D|}{|D|} = \dfrac{3}{4}$.

19	In a throw of a fair six-sided die, given that a prime number comes up, what is the probability that the number is at most 3?	2	L3	CO2

Prime outcomes: $\{2, 3, 5\}$; those at most 3: $\{2, 3\}$. So $P(X \leq 3 \mid \text{prime}) = 2/3$.

20	For a partition A, B, C of a sample space S and an event Y in S, write $P(A \mid Y)$ using Bayes' theorem.	1	L1	CO2

$$P(A \mid Y) = \frac{P(Y \mid A)P(A)}{P(Y \mid A)P(A) + P(Y \mid B)P(B) + P(Y \mid C)P(C)}.$$

21	Is the following a valid probability distribution? Why or why not? x: 1 / 2 / 3 / 4 $P(x)$: 4/7 / 3/7 / 2/7 / 1/7	2	L4	CO2

Every value lies between 0 and 1, but $\sum P(x) = \dfrac{4+3+2+1}{7} = \dfrac{10}{7} > 1$.

Answer: No; the probabilities do not add up to 1.

22	In a throw of a fair six-sided die, given that a non-prime number comes up, what is the probability that the number is odd?	2	L3	CO2

Non-prime outcomes: $\{1, 4, 6\}$; the only odd one is 1. So $P(\text{odd} \mid \text{non-prime}) = 1/3$.

23	State and explain the addition rule for two events in probability.	2	L1	CO2

For any two events A and B,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

$P(A \cap B)$ is subtracted because outcomes in both events are otherwise counted twice. If A and B are mutually exclusive, $P(A \cap B) = 0$ and $P(A \cup B) = P(A) + P(B)$.

24	What is conditional probability? Provide its formula.	2	L1	CO2

The conditional probability of A given B is the probability that A occurs when it is known that B has occurred:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0.$$

The sample space is effectively reduced to B.

25	State Bayes' theorem.	2	L1	CO2

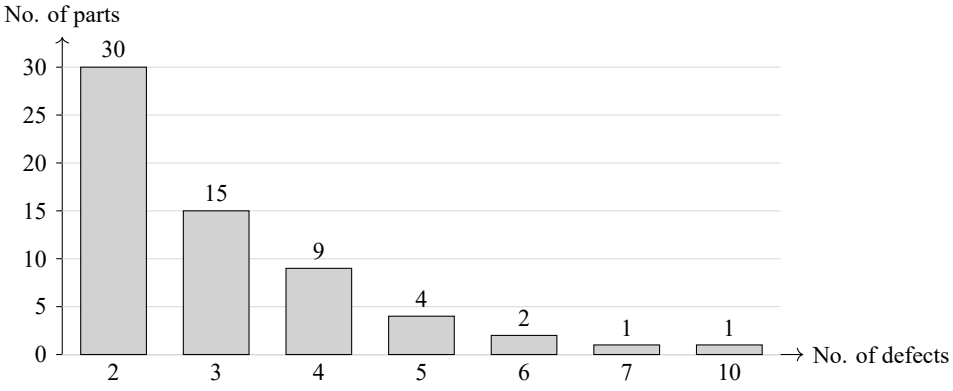
If $E_1, E_2, \ldots, E_k$ are mutually exclusive and exhaustive events and B is any event with $P(B) > 0$, then

$$P(E_i \mid B) = \frac{P(B \mid E_i)\, P(E_i)}{\sum_{j=1}^{k} P(B \mid E_j)\, P(E_j)}.$$

It updates the prior probability $P(E_i)$ of a cause to the posterior $P(E_i \mid B)$ once B is observed.

Sl. No.	Questions	M	BT	CO
	PART – B (Descriptive and numerical questions)			
1	62 parts were analysed for the number of defects with the following distribution: No. of defects, i: 2, 3, 4, 5, 6, 7, 10 No. of parts with i defects: 30, 15, 9, 4, 2, 1, 1 Plot a bar chart of the number of defects against the number of parts. Compute the mean, median, mode and standard deviation of the number of defects per part.	10	L3	CO1

Bar chart



Summary measures

$N = \sum f_i = 62, \sum f_i x_i = 60{+}45{+}36{+}20{+}12{+}7{+}10 = 190, \sum f_i x_i^2 = 120{+}135{+}144{+}100{+}72{+}49{+}100 = 720.$

$$\bar{x} = \frac{190}{62} = 3.0645$$

$$s^2 = \frac{\sum f_i x_i^2 - (\sum f_i x_i)^2/N}{N-1} = \frac{720 - 190^2/62}{61} = 2.2581, \qquad s = 1.5027$$

Median: the 31st and 32nd values both have 3 defects (cumulative frequencies 30, 45, …), so the median is 3. Mode: 2 defects (frequency 30).

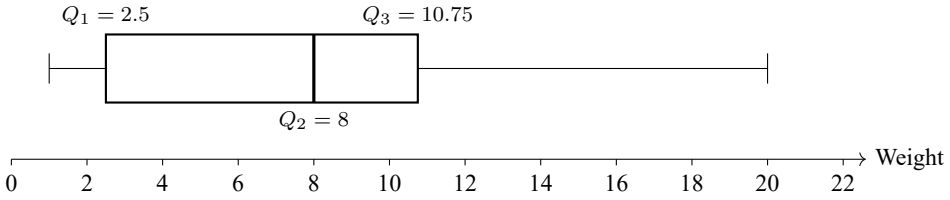
Answer: $\bar{x} = 3.065$, median $= 3$, mode $= 2$, $s = 1.503$ defects per part.

2	20 weight measurements in a process were recorded as: 2, 8, 11, 10, 8, 4, 2, 2, 9, 8, 4, 11, 12, 2, 4, 4, 20, 1, 10, 15. (i) Draw a box plot for the data. (ii) What are the range and the IQR of the data? (iii) Are there any outliers? If yes, what are they?	10	L3	CO1

Sorted: 1, 2, 2, 2, 2, 4, 4, 4, 4, 8, 8, 8, 9, 10, 10, 11, 11, 12, 15, 20 ($n = 20$).

$$Q_1 = x_{(5.25)} = 2 + 0.25(4 - 2) = 2.5, \quad Q_2 = \tfrac{8+8}{2} = 8, \quad Q_3 = x_{(15.75)} = 10 + 0.75(11 - 10) = 10.75$$

(i) Box plot (whiskers to the furthest points inside the fences):



(ii) Range $= 20 - 1 = 19$; IQR $= Q_3 - Q_1 = 8.25$.

(iii) Fences: $Q_1 - 1.5\,\text{IQR} = -9.875$ and $Q_3 + 1.5\,\text{IQR} = 23.125$. No value lies outside the fences, so there are no outliers.

Answer: range $= 19$, IQR $= 8.25$; outlier: none.

3	The nine measurements that follow are furnace temperatures (°F) recorded on successive batches in a semiconductor manufacturing process: 953, 950, 948, 955, 951, 949, 957, 954, 955. (i) Calculate the sample mean, sample variance and standard deviation. (ii) Find the mode and median. How much could the largest temperature measurement increase without changing the median? (iii) Construct a box plot of the data, showing clearly the outlier bounds. Comment on the symmetry of the data.	10	L3	CO1

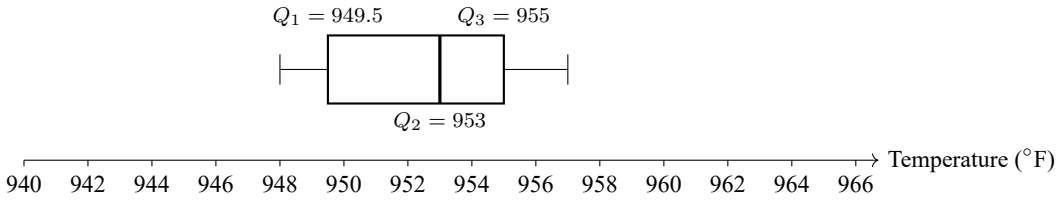
Sorted: 948, 949, 950, 951, 953, 954, 955, 955, 957 ($n = 9$).

(i) $\sum x = 8572$, so $\bar{x} = 8572/9 = 952.444$. $\sum x^2 = 8164430$,

$$s^2 = \frac{\sum x^2 - (\sum x)^2/n}{n-1} = \frac{8164430 - 8572^2/9}{8} = 9.5278, \qquad s = 3.0867°\text{F}.$$

(ii) Mode $= 955$ (occurs twice). Median $= x_{(5)} = 953$. The median depends only on the middle value, so the largest value (957) can increase by any amount without changing the median.

(iii) $Q_1 = x_{(2.5)} = 949.5$, $Q_3 = x_{(7.5)} = 955$, IQR $= 5.5$. Outlier bounds: $949.5 - 1.5(5.5) = 941.25$ and $955 + 1.5(5.5) = 963.25$; no value lies outside them.



The median sits slightly right of the centre of the box and the whiskers are of similar length, so the data are roughly symmetric with a mild left skew (mean $952.44 <$ median 953).

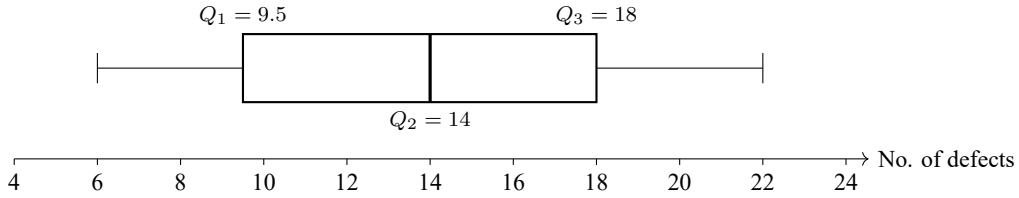
Answer: $\bar{x} = 952.444$, $s^2 = 9.528$, $s = 3.087$; mode 955, median 953; no outliers.

4	The numbers of defects when 25 parts were analysed were: 15, 18, 20, 9, 11, 8, 16, 14, 13, 15, 17, 21, 7, 9, 11, 8, 18, 12, 6, 19, 15, 22, 19, 12, 10. (i) Plot a stem-and-leaf diagram of the number of defects in a part. (ii) Draw a box plot for the data and comment on the distribution.	08	L3	CO1

(i) Stem = tens digit, leaf = units digit:

Stem	Leaf	Freq.
0	6 7 8 8 9 9	6
1	0 1 1 2 2 3 4 5 5 5 6 7 8 8 9 9	16
2	0 1 2	3

(ii) $n = 25$: $Q_1 = x_{(6.5)} = 9.5$, $Q_2 = x_{(13)} = 14$, $Q_3 = x_{(19.5)} = 18$, IQR $= 8.5$; fences -3.25 and 30.75, so there are no outliers.

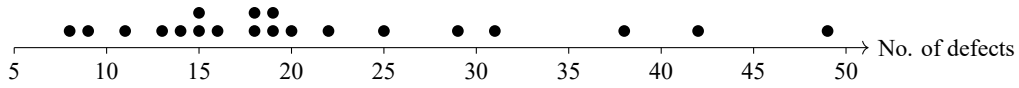


The median is near the middle of the box and the whiskers are about equal, so the distribution is fairly symmetric. The values range from 6 to 22, with the middle half between 9.5 and 18 defects.

5	The numbers of defects when 20 parts were analysed were: 15, 18, 20, 29, 31, 8, 16, 14, 13, 25, 49, 11, 38, 18, 42, 9, 19, 15, 22, 19. Draw the following plots and comment on the data using each plot: (i) dot plot, (ii) stem-and-leaf diagram, (iii) box plot.	10	L3	CO1

Sorted: 8, 9, 11, 13, 14, 15, 15, 16, 18, 18, 19, 19, 20, 22, 25, 29, 31, 38, 42, 49.

(i) Dot plot:



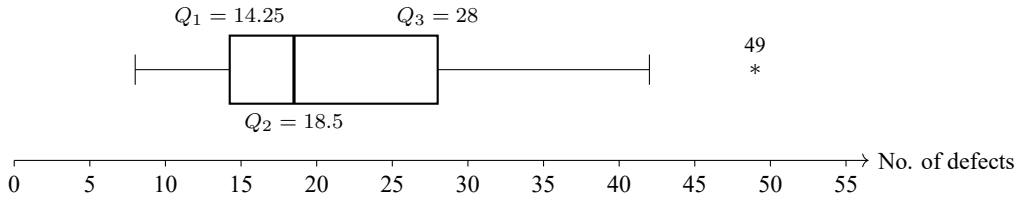
Most parts have 8–31 defects, clustered around 13–22; the values 38, 42 and 49 stand apart in a long right tail.

(ii) Stem-and-leaf (stem = tens):

Stem	Leaf	Freq.
0	8 9	2
1	1 3 4 5 5 6 8 8 9 9	10
2	0 2 5 9	4
3	1 8	2
4	2 9	2

Half of the values are in the teens; the leaves thin out upwards, showing right skew.

(iii) $Q_1 = x_{(5.25)} = 14.25$, $Q_2 = 18.5$, $Q_3 = x_{(15.75)} = 28$, IQR $= 13.75$; fences -6.375 and 48.625.



The value 49 lies beyond the upper fence and is an outlier. The longer upper whisker and box half confirm the right skew.

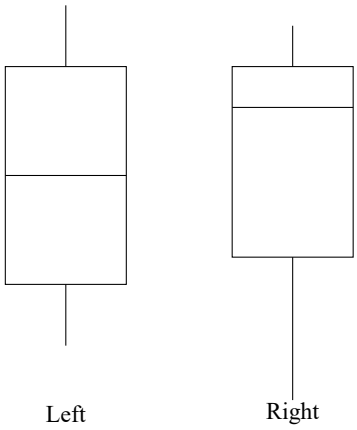
6	Differentiate between nominal, ordinal, interval and ratio data types with examples.	06	L2	CO1

Type	**Description**	**Examples**
Nominal	Categories with names only; no order. Only counting and the mode are meaningful.	Gender, blood group, machine ID
Ordinal	Categories with a natural order, but the gaps between them are not equal or known.	Satisfaction rating, grade A/B/C, defect severity
Interval	Numeric with equal intervals but no true zero; differences are meaningful, ratios are not.	Temperature in °C, calendar year
Ratio	Numeric with equal intervals and a true zero; ratios are meaningful.	Length, weight, time, number of defects

7	What are the key components of a radar chart? Where is it used?	06	L1	CO1

- **Axes:** one axis (spoke) for each variable, radiating from a common centre at equal angles.
- **Scale:** a common scale along each spoke, usually with concentric grid polygons or circles.
- **Data polygon:** the value of each variable is plotted on its spoke and the points are joined to form a polygon; several items can be overlaid.
- **Labels and legend:** variable names at the spoke ends and a legend for the items compared.

Uses: comparing several items on many attributes at once, such as employee skill profiles, supplier or product performance against several criteria, quality-attribute scorecards, and benchmarking. Large or lopsided polygons show strengths and weaknesses at a glance.

8	Describe any similarities or differences in the two distributions represented by the following box plots. Assume the two box plots have the same scale. 	06	L4	CO1

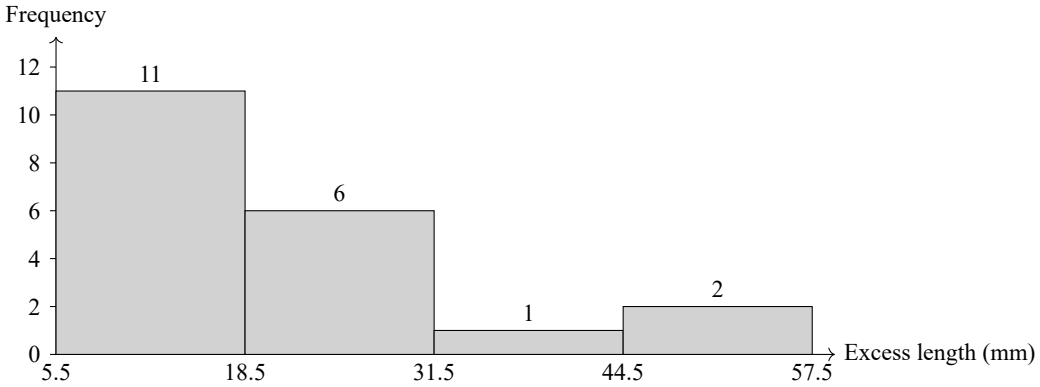
- **Similarities:** the boxes have about the same length (similar IQR) and the upper ends of the boxes (Q_3) are at about the same level, so the spread of the middle half and the upper quartile are similar.
- **Centre:** the left plot has its median in the middle of the box; the right plot has its median close to Q_3, so its median is higher.
- **Shape:** the left distribution is symmetric (equal box halves and whiskers). The right one is negatively (left) skewed: the lower half of the box and the lower whisker are much longer than the upper ones.
- **Range:** the right plot extends further at the bottom, so its minimum is lower and its overall range is larger.

9	The excess length (mm) of a manufactured part is recorded as: 19, 22, 6, 29, 57, 14, 12, 24, 6, 37, 19, 19, 18, 12, 11, 17, 15, 17, 13, 56. (i) Compute the mean, median and mode of the data. (ii) Draw a histogram with 4 classes. (iii) Draw a box plot and interpret it.	10	L3	CO1

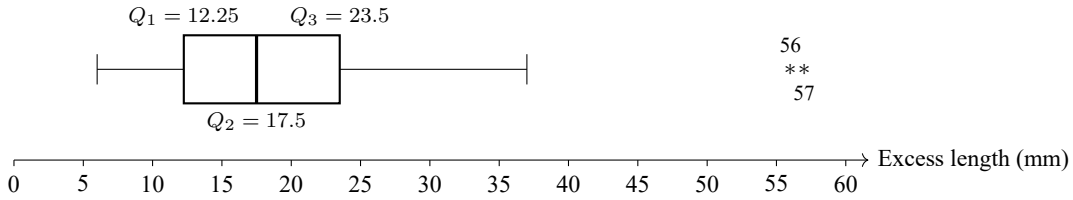
Sorted: 6, 6, 11, 12, 12, 13, 14, 15, 17, 17, 18, 19, 19, 19, 22, 24, 29, 37, 56, 57 ($n = 20$).

(i) $\sum x = 423$, $\bar{x} = 423/20 = 21.15$ mm. Median $= \frac{x_{(10)}+x_{(11)}}{2} = \frac{17+18}{2} = 17.5$ mm. Mode $= 19$ mm (occurs 3 times).

(ii) Range $= 57 - 6 = 51$; with 4 classes the width is $51/4 \approx 13$. Classes 6–18, 19–31, 32–44, 45–57 (boundaries 5.5, 18.5, 31.5, 44.5, 57.5) have frequencies 11, 6, 1, 2.



(iii) $Q_1 = x_{(5.25)} = 12.25$, $Q_3 = x_{(15.75)} = 23.5$, IQR $= 11.25$; upper fence $= 40.375$.



The values 56 and 57 lie beyond the upper fence and are outliers. The distribution is right-skewed: the mean (21.15) is well above the median (17.5).

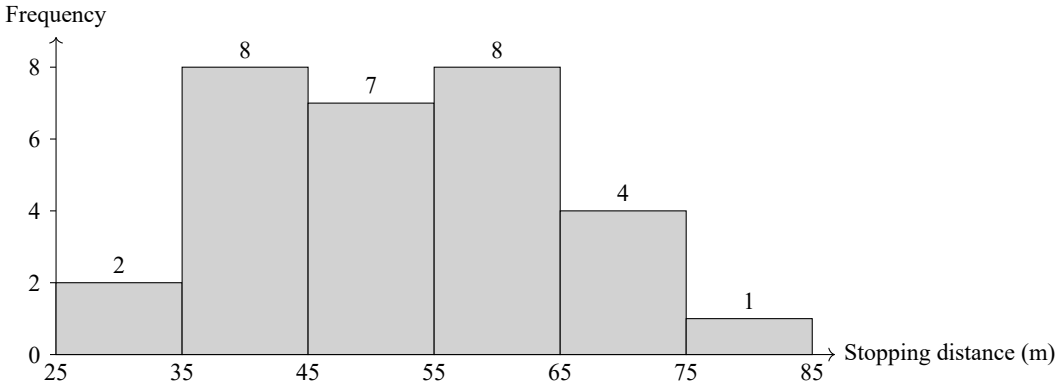
Answer: mean $= 21.15$ mm, median $= 17.5$ mm, mode $= 19$ mm; outliers 56 and 57.

10	Experiments were conducted to study the braking system of a standard car. At 45 km/h the brakes were applied and the stopping distances (m) were: 76, 70, 63, 53, 55, 62, 47, 30, 64, 42, 39, 56, 43, 66, 44, 48, 70, 71, 45, 47, 40, 41, 29, 42, 58, 44, 51, 60, 62, 49. (i) Tabulate the frequency distribution. (ii) Plot the frequency histogram. (iii) Compute the mean, median and standard deviation. (iv) Draw the stem-and-leaf diagram.	10	L3	CO1

(i) Classes of width 10 (lower limit included):

Distance (m)	25–35	35–45	45–55	55–65	65–75	75–85
Frequency	2	8	7	8	4	1

(ii) Histogram:



(iii) $n = 30$, $\sum x = 1567$, $\sum x^2 = 86201$. $\bar{x} = 52.233$ m; median $= \frac{x_{(15)}+x_{(16)}}{2} = 50$ m; $s = \sqrt{\dfrac{\sum x^2 - (\sum x)^2/n}{n-1}} = \sqrt{150.047} = 12.249$ m.

(iv) Stem = tens, leaf = units:

Stem	Leaf	Freq.
2	9	1
3	0 9	2
4	0 1 2 2 3 4 4 5 7 7 8 9	12
5	1 3 5 6 8	5
6	0 2 2 3 4 6	6
7	0 0 1 6	4

Most stopping distances lie between 40 and 65 m; the distribution is slightly right-skewed and has no extreme values.

Answer: $\bar{x} = 52.23$ m, median $= 50$ m, $s = 12.25$ m.

11	Compare and contrast histograms, box plots and stem-and-leaf plots in terms of their use and the information they provide.	06	L2	CO1

Feature	Histogram	Box plot	Stem-and-leaf
Shows	Frequency of each class; overall shape, modality, spread	Five-number summary (min, Q_1, median, Q_3, max) and outliers	Shape plus every original value
Data size	Medium to large data sets	Any size; best for comparing groups	Small data sets (up to about 50 values)
Raw values	Lost (grouped into classes)	Lost (only summaries)	Retained
Skewness, outliers	Seen from the shape; outliers not flagged	Skew from the median position; outliers flagged explicitly	Seen from the leaf lengths
Typical use	Distribution of a process variable	Comparing machines, shifts or suppliers side by side	Quick exploratory display by hand

All three display the distribution of one numerical variable. A histogram is best for the shape of large data, a box plot for comparison and outlier detection, and a stem-and-leaf plot keeps the raw data for small samples.

12	What are stem-and-leaf diagrams? Illustrate with an example and mention two applications.	06	L2	CO1

A stem-and-leaf diagram displays numerical data by splitting each value into a **stem** (the leading digit or digits) and a **leaf** (the last digit). Stems are written in a column and the leaves of each stem are listed in order beside it, so the display looks like a histogram turned on its side while every value can still be read.

Example: the data 12, 15, 21, 22, 29, 34, 36, 38 give

Stem	Leaf
1	2 5
2	1 2 9
3	4 6 8

Applications: (1) a quick display of the shape, centre and spread of a small data set, for example test scores or daily defect counts; (2) reading the median, quartiles and mode directly when the original values are needed.

13	The lives (to the nearest tenth of a year) of 36 batteries are: 4.1, 5.2, 2.8, 4.9, 5.6, 4.0, 4.1, 4.3, 5.4, 4.5, 6.1, 3.7, 2.3, 4.5, 4.9, 5.6, 4.3, 3.9, 3.2, 5.0, 4.8, 3.7, 4.6, 5.5, 1.8, 5.1, 4.2, 6.3, 3.3, 5.8, 4.4, 4.8, 3.0, 4.3, 4.7, 5.1. Construct a stem-and-leaf plot and comment on it.	08	L3	CO1

Stem = whole years, leaf = tenths of a year:

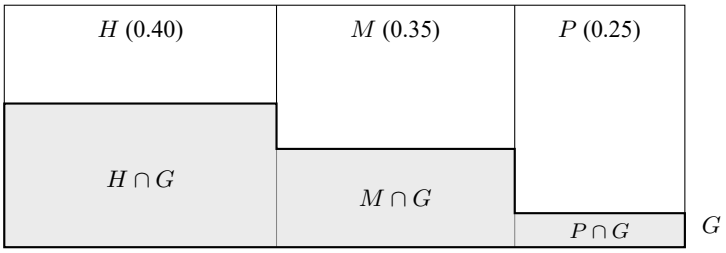
Stem	Leaf	Freq.
1	8	1
2	3 8	2
3	0 2 3 7 7 9	6
4	0 1 1 2 3 3 3 4 5 5 6 7 8 8 9 9	16
5	0 1 1 2 4 5 6 6 8	9
6	1 3	2

The distribution is single-peaked and roughly symmetric, centred near 4.5 years (median $= 4.5$, mean $= 4.439$), with most batteries lasting 3 to 6 years; 1.8 years is a low value.

14	Customers evaluate products and either give them good reviews or do not. Products are also highly successful, moderately successful or poor according to sales. In the past, 95% of highly successful products received good reviews, 60% of moderately successful products received good reviews, and 10% of poor products received good reviews. 40% of products have been highly successful, 35% moderately successful, and the rest poor. (i) Depict the sample space with appropriate partitions in a sketch. What is the probability that a product attains a good review? (ii) If a product attains a good review, what is the probability that it is a highly successful product? (iii) If a product does not attain a good review, what is the probability that it is a highly successful product?	08	L3	CO2

Let H, M, P be highly, moderately and poorly successful ($P(H) = 0.40$, $P(M) = 0.35$, $P(P) = 0.25$) and G a good review.

(i) The sample space is partitioned by H, M, P, and G cuts across all three:



By the total probability rule,

$$P(G) = 0.40(0.95) + 0.35(0.60) + 0.25(0.10) = 0.38 + 0.21 + 0.025 = 0.615.$$

(ii) $P(H \mid G) = \dfrac{0.40(0.95)}{0.615} = \dfrac{0.38}{0.615} = 0.6179.$

(iii) $P(G') = 1 - 0.615 = 0.385$ and $P(H \cap G') = 0.40(0.05) = 0.02$, so $P(H \mid G') = \dfrac{0.02}{0.385} = 0.0519.$

15	Consider the following random experiments. Define the sample space for each and identify the type of sample space. (i) An observer conducting a traffic survey counts the traffic offences during peak hours. (ii) A chemical analyst records the pH value of water using a pH meter. (iii) An automobile manufacturer counts the model variants created from three capacities (800 cc, 1000 cc, 1500 cc), two options (with AC, without AC), three colours (steel grey, beige, peach) and three fuel systems (petrol, diesel, hybrid). (iv) The time until a transaction service is requested of a computer. (v) The number of erroneous tracks in a digital recording tape in which each of 24 tracks is classified as containing or not containing one or more errors.	08	L2	CO2

	Sample space	**Type**
(i)	$S = \{0, 1, 2, \dots\}$	Discrete, countably infinite
(ii)	$S = \{x : 0 \le x \le 14\}$	Continuous
(iii)	All combinations (capacity, AC, colour, fuel); $3 \times 2 \times 3 \times 3 = 54$ outcomes	Discrete, finite
(iv)	$S = \{t : t > 0\}$ (or $t \ge 0$)	Continuous
(v)	$S = \{0, 1, 2, \dots, 24\}$	Discrete, finite

16	Pouches of a specialty chemical are tested for impurities A and B, either of which makes the chemical unfit for use. Suppose 10% of the pouches have impurity A, 5% have impurity B, 1% have both, and the others have no impurities. Pouches are unfit for use with probabilities 1%, 2%, 6% and 0.1% if they have impurity A, impurity B, both impurities, and no impurities, respectively. What is the probability that a randomly chosen pouch is unfit for use?	08	L3	CO2

The four impurity states must be mutually exclusive: A only, B only, both, none.

$P(\text{A only}) = 0.10 - 0.01 = 0.09, \quad P(\text{B only}) = 0.05 - 0.01 = 0.04, \quad P(\text{both}) = 0.01, \quad P(\text{none}) = 1 - 0.14 = 0.86.$

By the total probability rule,

$$P(U) = 0.09(0.01) + 0.04(0.02) + 0.01(0.06) + 0.86(0.001) = 0.0009 + 0.0008 + 0.0006 + 0.00086 = 0.00316.$$

Answer: $P(\text{unfit}) = 0.00316$ (about 0.32%).

17	Disks of polycarbonate plastic from a supplier are analysed for scratch and shock resistance. The results from 100 disks are: | | **Shock resistance high** | **Shock resistance low** | | Scratch resistance high | 70 | 9 | | Scratch resistance low | 16 | 5 | Let A denote the event that a disk has high shock resistance and B the event that it has high scratch resistance. For a disk selected at random, find (a) $P(A)$ (b) $P(B)$ (c) $P(A')$ (d) $P(A \cap B)$ (e) $P(A \cup B)$ (f) $P(A' \cup B)$ (g) $P(A \mid B)$ (h) $P(B \mid A)$.	08	L3	CO2

Row and column totals: scratch high 79, scratch low 21, shock high 86, shock low 14.

(a) $P(A) = 86/100 = 0.86$ (b) $P(B) = 79/100 = 0.79$ (c) $P(A') = 1 - 0.86 = 0.14$ (d) $P(A \cap B) = 70/100 = 0.70$

(e) $P(A \cup B) = 0.86 + 0.79 - 0.70 = 0.95$

(f) $A' \cup B$ excludes only the disks with high shock and low scratch resistance: $P(A' \cup B) = 1 - 16/100 = 0.84$

(g) $P(A \mid B) = 70/79 = 0.8861$ (h) $P(B \mid A) = 70/86 = 0.814$

18	In a factory, machines 1, 2 and 3 all produce pins. Machines 1, 2 and 3 produce 2%, 1% and 3% defective pins respectively, and they make 35%, 25% and 40% of the total production. If one pin is selected at random from the total output, what is the probability that it is defective? Find also the conditional probability that it was produced by machine 3, given that it is defective.	08	L3	CO2

Let M_i be "made on machine i" and D "defective". By the total probability rule,

$$P(D) = 0.35(0.02) + 0.25(0.01) + 0.40(0.03) = 0.007 + 0.0025 + 0.012 = 0.0215.$$

By Bayes' theorem,

$$P(M_3 \mid D) = \frac{0.40(0.03)}{0.0215} = \frac{0.012}{0.0215} = 0.5581.$$

Answer: $P(D) = 0.0215$; $P(M_3 \mid D) = 0.5581$.

19	The rise time of a reactor is measured in minutes (and fractions of minutes). Let the sample space be the positive real numbers, and define the events $A = \{x \mid x < 72.5\}$ and $B = \{x \mid x > 52.5\}$. Describe each of the following events: (i) A' (ii) B' (iii) $A \cap B$ (iv) $A \cup B$.	08	L2	CO2

(i) $A' = \{x \mid x \geq 72.5\}$: rise time of at least 72.5 min.

(ii) $B' = \{x \mid 0 < x \leq 52.5\}$: rise time of at most 52.5 min.

(iii) $A \cap B = \{x \mid 52.5 < x < 72.5\}$: rise time strictly between 52.5 and 72.5 min.

(iv) $A \cup B = \{x \mid x > 0\}$: every positive rise time, i.e. the whole sample space S (every x is either below 72.5 or above 52.5).

20	It is known that 1% of a population suffers from a particular disease. A blood test has a 97% chance of a positive result for diseased individuals, but also a 6% chance of falsely indicating that a healthy person has the disease. (i) What is the probability that a person will have a positive blood test? (ii) If your blood test is positive, what is the chance that you have the disease?	08	L3	CO2

Let D = has the disease and T = positive test: $P(D) = 0.01$, $P(T \mid D) = 0.97$, $P(T \mid D') = 0.06$.

(i) $P(T) = 0.01(0.97) + 0.99(0.06) = 0.0097 + 0.0594 = 0.0691$.

(ii) $P(D \mid T) = \dfrac{0.0097}{0.0691} = 0.1404$, i.e. only about 14%: because the disease is rare, most positive results come from healthy people.

21	100 mobile screens are inspected for touch sensitivity and display quality, with the results below. Let A = high touch sensitivity and B = high display quality. Find $P(A)$, $P(B)$, $P(A')$, $P(A \cap B)$, $P(A \cup B)$, $P(A' \cup B)$, $P(A \mid B)$ and $P(B \mid A)$.	08	L3	CO2

	Touch high	**Touch low**
Display high	68	12
Display low	14	6

Totals: touch high 82, touch low 18, display high 80, display low 20.

- $P(A) = 0.82$, $P(B) = 0.80$, $P(A') = 0.18$, $P(A \cap B) = 0.68$
- $P(A \cup B) = 0.82 + 0.80 - 0.68 = 0.94$; $\quad P(A' \cup B) = 1 - P(A \cap B') = 1 - 0.14 = 0.86$
- $P(A \mid B) = 68/80 = 0.85$; $\quad P(B \mid A) = 68/82 = 0.8293$

UNIT II – RANDOM VARIABLES AND DISCRETE PROBABILITY DISTRIBUTIONS

Discrete and continuous random variables; probability mass and density functions; cumulative distribution functions; discrete uniform, binomial, Poisson, geometric and negative binomial distributions

Sl. No.	Questions	M	BT	CO
	PART – A (Short-answer questions)			
1	A table listing all possible values of a random variable and their corresponding probabilities is called _____.	1	L1	CO2
	A probability distribution (for a discrete random variable, its probability mass function).			
2	If $f(x)$ is the probability density function of a continuous random variable, what is $\int_{-\infty}^{\infty} f(x)\,dx$?	1	L1	CO2
	$\int_{-\infty}^{\infty} f(x)\,dx = 1$, because the total probability over the whole range of the variable is 1.			
3	The probability that a patient is successfully treated is $p = 0.80$. If the treatment is used on 40 patients, what is the expected number of patients successfully treated?	2	L3	CO2
	The number treated successfully is binomial with $n = 40$, $p = 0.8$, so $E(X) = np = 40(0.80) = 32$ patients.			
4	The sum of all the probabilities in a probability mass function equals _____.	1	L1	CO2
	One: $\sum_x P(X = x) = 1$.			
5	Which function gives the probability that a discrete random variable equals a specific value?	1	L1	CO2
	The probability mass function, $f(x) = P(X = x)$.			
6	Differentiate between discrete and continuous probability distributions.	2	L2	CO2
	A discrete distribution is defined on a finite or countable set of values and is described by a probability mass function, with $P(X = x) > 0$ at those values (binomial, Poisson). A continuous distribution is defined on an interval and is described by a probability density function; probabilities are areas under the curve and $P(X = x) = 0$ for any single value (normal, exponential).			
7	Differentiate between discrete and continuous random variables with examples.	2	L2	CO2
	A discrete random variable takes a finite or countable set of values, for example the number of defective parts in a lot or the number of calls per hour. A continuous random variable can take any value in an interval, for example the length of a part, the time to failure, or a temperature.			

8	For a continuous random variable, the probability that it takes an exact value is _____.	1	L1	CO2

Zero: $P(X = a) = \int_a^a f(x)\,dx = 0$ for every single value a.

9	The area under the probability density function of a continuous random variable over its entire range is _____.	1	L1	CO2

Equal to 1.

10	A function $f(x)$ is a valid probability mass function if $\sum f(x) =$ _____ and $f(x) \geq$ _____ for all x.	2	L1	CO2

$\sum f(x) = 1$ and $f(x) \geq 0$ for all x.

11	Give an example of a situation that leads to a discrete probability distribution.	2	L2	CO2

The number of defective items in a sample of 20 parts taken from a production line: it can take only the values $0, 1, \ldots, 20$, so it has a discrete (binomial) distribution. Other examples: the number of customers arriving in an hour, or the face shown by a die.

12	What are the properties of a probability mass function (PMF)?	2	L1	CO2

For a discrete random variable X with possible values $x_1, x_2, \ldots$: (1) $f(x_i) \geq 0$ for all x_i; (2) $\sum_i f(x_i) = 1$; (3) $f(x_i) = P(X = x_i)$, and the probability of any set of values is the sum of f over that set.

13	The binomial distribution models the number of _____ in a fixed number of independent trials.	1	L1	CO2

Successes.

14	In a geometric distribution, the random variable represents the number of trials until the first _____.	1	L1	CO2

Success.

15	In the negative binomial distribution, the random variable counts the number of trials needed to achieve _____ successes.	1	L1	CO2

A specified number r of successes.

16	Mention any two characteristics of the binomial distribution.	2	L1	CO2

(1) There is a fixed number n of independent trials, each with only two outcomes (success or failure). (2) The probability of success p is the same in every trial. Its mean is np and its variance $np(1-p)$.

17	Cyclists arrive at a summit as a Poisson process with intensity $\lambda = 8$ per hour. Given that no cyclist arrived in the last half hour, find the probability that exactly 3 cyclists arrive in the next half hour.	2	L3	CO2

Counts in disjoint intervals of a Poisson process are independent, so the past half hour does not matter. For half an hour the mean is $8 \times \frac{1}{2} = 4$:

$$P(X = 3) = \frac{e^{-4}4^3}{3!} = 0.1954.$$

18	Calls at a call centre follow a Poisson distribution with a mean of 4 per hour. Find the probability that exactly 2 calls are received in an hour.	2	L3	CO2

$$P(X = 2) = \frac{e^{-4}4^2}{2!} = 8e^{-4} = 0.1465.$$

19	Accidents occur as a Poisson process with a rate of 3 per day. Find the probability of no accidents on a given day.	2	L3	CO2

$P(X = 0) = e^{-3} = 0.0498.$

20	If the probability of success is 0.2, find the probability that the first success occurs on the 4th trial.	2	L3	CO2

Geometric distribution: $P(X = 4) = (1-p)^3 p = (0.8)^3(0.2) = 0.1024.$

21	A negative binomial random variable counts the trials needed to get 3 successes with $p = 0.4$. Find the probability that the 3rd success occurs on the 5th trial.	2	L3	CO2

$$P(X = 5) = \binom{5-1}{3-1} p^3(1-p)^2 = \binom{4}{2}(0.4)^3(0.6)^2 = 6(0.064)(0.36) = 0.1382.$$

Sl. No.	Questions	M	BT	CO
	PART – B (Descriptive and numerical questions)			
1	The probability density function of the weight of packages delivered by a post office is $f(x) = \frac{70}{69x^2}$ for $1 < x < 70$ pounds. (i) What is the probability that a package weighs less than 10 pounds? (ii) Determine the mean and variance of package weight. (iii) If the shipping cost is \$3 per pound, what is the average shipping cost of a package?	10	L3	CO2

(i) $P(X < 10) = \int_1^{10} \frac{70}{69x^2}\,dx = \frac{70}{69}\Big[-\frac{1}{x}\Big]_1^{10} = \frac{70}{69}\Big(1 - \frac{1}{10}\Big) = 0.913.$

(ii) $E(X) = \int_1^{70} x \cdot \frac{70}{69x^2}\,dx = \frac{70}{69}\ln 70 = 4.3101$ lb; $E(X^2) = \int_1^{70} \frac{70}{69}\,dx = \frac{70}{69}(69) = 70$, so $V(X) = 70 - 4.3101^2 = 51.423$ lb^2.

(iii) Cost $= 3X$, so $E(3X) = 3E(X) = 3(4.3101) = \$12.93$.

2	A random variable X has the cumulative distribution function $$F(x) = \begin{cases} 0, & x < 1 \\ 0.4, & 1 \le x < 3 \\ 0.5, & 3 \le x < 8 \\ 1.0, & 8 \le x \end{cases}$$ Determine (i) $P(X \le 3)$ (ii) $E(X)$ (iii) $P(1 \le X \le 2)$ (iv) $P(X \ge 2)$ (v) the median of the distribution.	08	L3	CO2

The jumps of F give the PMF: $P(X = 1) = 0.4$, $P(X = 3) = 0.5 - 0.4 = 0.1$, $P(X = 8) = 1 - 0.5 = 0.5$.

(i) $P(X \le 3) = F(3) = 0.5$.

(ii) $E(X) = 1(0.4) + 3(0.1) + 8(0.5) = 0.4 + 0.3 + 4.0 = 4.7$.

(iii) $P(1 \le X \le 2) = P(X = 1) = 0.4$ (there is no probability at 2).

(iv) $P(X \ge 2) = 1 - P(X < 2) = 1 - F(1) = 0.6$.

(v) The median is the smallest x with $F(x) \ge 0.5$, which is $x = 3$. (Since $F(x) = 0.5$ exactly for $3 \le x < 8$, any value in $[3, 8]$ splits the probability in half; 3 is the conventional choice.)

3	A discrete random variable X has the probability mass function x: 1, 2, 3, 4 $P(X = x)$: 0.1, 0.2, 0.4, 0.3 (a) Find the cumulative distribution function (CDF) of X. (b) Calculate $E(X)$ and $V(X)$.	10	L3	CO2

(a)

$$F(x) = \begin{cases} 0, & x < 1 \\ 0.1, & 1 \le x < 2 \\ 0.3, & 2 \le x < 3 \\ 0.7, & 3 \le x < 4 \\ 1, & x \ge 4 \end{cases}$$

(b) $E(X) = 1(0.1) + 2(0.2) + 3(0.4) + 4(0.3) = 2.9$; $E(X^2) = 1(0.1) + 4(0.2) + 9(0.4) + 16(0.3) = 9.3$; $V(X) = 9.3 - 2.9^2 = 0.89$.

4	Marketing estimates that a new instrument for the analysis of soil samples will be very successful, moderately successful or unsuccessful, with probabilities 0.3, 0.6 and 0.1 respectively. The yearly revenue for a very successful, moderately successful or unsuccessful product is \$10 million, \$5 million and \$1 million respectively. Let X denote the yearly revenue of the product. Determine the probability mass function of X, and compute its mean and variance.	10	L3	CO2

PMF (in \$ million):

x	10	5	1
$f(x)$	0.3	0.6	0.1

$$E(X) = 10(0.3) + 5(0.6) + 1(0.1) = 6.1, \qquad E(X^2) = 100(0.3) + 25(0.6) + 1(0.1) = 45.1$$

$$V(X) = 45.1 - 6.1^2 = 7.89.$$

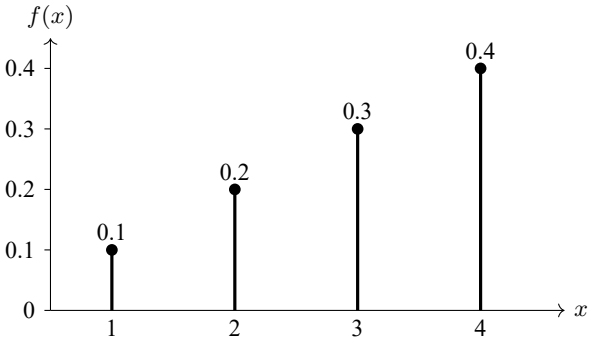
Answer: mean \$6.1 million; variance 7.89 (\$ million)^2, i.e. standard deviation \$2.809 million.

5	A random variable X is distributed as $f(x) = x/k$ with range $\{1, 2, 3, 4\}$. (i) What is the probability that X takes a value of at most 2? (ii) Plot the probability mass function of X. (iii) Derive the cumulative distribution function of X and plot it. (iv) Find the mean and variance of X.	10	L3	CO2

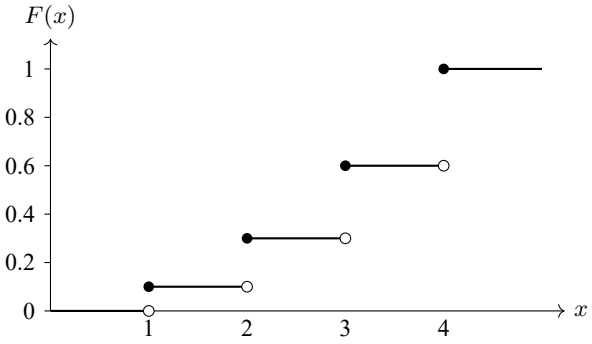
Since the probabilities sum to 1, $\frac{1+2+3+4}{k} = 1$, so $k = 10$ and $f(1) = 0.1$, $f(2) = 0.2$, $f(3) = 0.3$, $f(4) = 0.4$.

(i) $P(X \leq 2) = 0.1 + 0.2 = 0.3$.

(ii) PMF:



(iii) $F(x) = 0$ for $x < 1$; 0.1 for $1 \leq x < 2$; 0.3 for $2 \leq x < 3$; 0.6 for $3 \leq x < 4$; 1 for $x \geq 4$.



(iv) $E(X) = \frac{1+4+9+16}{10} = 3$; $E(X^2) = \frac{1+8+27+64}{10} = 10$; $V(X) = 10 - 9 = 1$.

6	A continuous random variable X has the probability density function $$f(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$ (i) Find $P(0.25 < X < 0.75)$. (ii) Find $E(X)$ and $V(X)$.	10	L3	CO2

(i) $P(0.25 < X < 0.75) = \int_{0.25}^{0.75} 2x\,dx = \left[x^2\right]_{0.25}^{0.75} = 0.5625 - 0.0625 = 0.5.$

(ii) $E(X) = \int_0^1 2x^2\,dx = \frac{2}{3} = 0.6667; \quad E(X^2) = \int_0^1 2x^3\,dx = \frac{1}{2}; \quad V(X) = \frac{1}{2} - \frac{4}{9} = \frac{1}{18} = 0.0556.$

7	The probability mass function of a discrete random variable X is $f(x) = \frac{x}{15}$, $x = 1, 2, 3, 4, 5$. (i) Find the expected value $E(X)$. (ii) Find the probability that X takes a value greater than 3.	08	L3	CO2

Check: $\sum f(x) = \frac{1+2+3+4+5}{15} = 1$.
(i) $E(X) = \sum x\,\frac{x}{15} = \frac{1+4+9+16+25}{15} = \frac{55}{15} = 3.6667$.
(ii) $P(X > 3) = f(4) + f(5) = \frac{4+5}{15} = \frac{9}{15} = 0.6$.

8	Past experience indicates that the ticket office at the Rex Theatre receives a mean of thirty calls per hour. Assuming the number of calls has a Poisson distribution, find the probability that (i) during a ten-minute coffee break, no calls are received; (ii) during a twenty-minute coffee break, at least two calls are received; (iii) during a half-hour lunch break, at most 3 calls are received.	10	L3	CO2

The rate is $30/60 = 0.5$ calls per minute.
(i) 10 min: $\lambda = 5$, $P(X = 0) = e^{-5} = 0.00674$.
(ii) 20 min: $\lambda = 10$, $P(X \geq 2) = 1 - e^{-10}(1 + 10) = 1 - 11e^{-10} = 0.9995$.
(iii) 30 min: $\lambda = 15$, $P(X \leq 3) = e^{-15}\left(1 + 15 + \frac{15^2}{2} + \frac{15^3}{6}\right) = 691e^{-15} = 0.000211$.

9	The number of flaws in bolts of cloth in textile manufacturing is assumed to be Poisson distributed with a mean of 0.1 flaw per square metre. (a) What is the probability that there are two flaws in 1 m^2 of cloth? (b) What is the probability that there is one flaw in 10 m^2 of cloth? (c) What is the probability that there are no flaws in 20 m^2 of cloth? (d) What is the probability that there are at least two flaws in 10 m^2 of cloth?	10	L3	CO2

(a) $\lambda = 0.1$: $P(X = 2) = \dfrac{e^{-0.1}(0.1)^2}{2!} = 0.00452$.
(b) $\lambda = 0.1 \times 10 = 1$: $P(X = 1) = e^{-1} = 0.3679$.
(c) $\lambda = 0.1 \times 20 = 2$: $P(X = 0) = e^{-2} = 0.1353$.
(d) $\lambda = 1$: $P(X \geq 2) = 1 - e^{-1}(1 + 1) = 1 - 2e^{-1} = 0.2642$.

10	Field observation shows that the traffic flow past a point on a highway segment is 720 vehicles per hour. Assume that vehicles arrive independently and follow a Poisson distribution. (i) Calculate the probability that no vehicle passes this point in 5 seconds. (ii) Calculate the probability that exactly 5 vehicles pass this point in 5 seconds. (iii) Calculate the probability that fewer than 5 vehicles pass this point in 5 seconds.	10	L3	CO2

Rate $= 720/3600 = 0.2$ vehicles per second, so in 5 s the mean is $\lambda = 1$.
(i) $P(X = 0) = e^{-1} = 0.3679$.
(ii) $P(X = 5) = \dfrac{e^{-1}1^5}{5!} = \dfrac{e^{-1}}{120} = 0.00307$.
(iii) $P(X < 5) = e^{-1}\left(1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24}\right) = 0.9963$.

11	8% of LED bulbs are defective. Inspecting bulbs independently, find the probability of (i) no defective in a sample of 12; (ii) at least one defective in a sample of 25; (iii) exactly two defective in a sample of 15.	10	L3	CO2

The number of defectives is binomial with $p = 0.08$.

(i) $n = 12$: $P(X = 0) = (0.92)^{12} = 0.3677$.

(ii) $n = 25$: $P(X \geq 1) = 1 - (0.92)^{25} = 1 - 0.1244 = 0.8756$.

(iii) $n = 15$: $P(X = 2) = \binom{15}{2}(0.08)^2(0.92)^{13} = 105(0.0064)(0.3383) = 0.2273$.

12	A carton of washers contains 5% whose variability in thickness around the circumference is unacceptable. (i) What is the probability that none of the unacceptable washers is found in a sample of 10 washers? (ii) What is the probability that at least one unacceptable washer is found in a sample of 20 washers?	10	L3	CO2

The number of unacceptable washers in a sample is binomial with $p = 0.05$ (a large carton).

(i) $n = 10$: $P(X = 0) = (0.95)^{10} = 0.5987$.

(ii) $n = 20$: $P(X \geq 1) = 1 - (0.95)^{20} = 1 - 0.3585 = 0.6415$.

13	Suppose X has a Poisson distribution with a mean of 4. Determine (i) $P(X = 0)$ (ii) $P(X = 8)$ (iii) $P(X \leq 2)$ (iv) $P(X = 4)$.	10	L3	CO2

$P(X = x) = \dfrac{e^{-4}4^x}{x!}$.

(i) $P(X = 0) = e^{-4} = 0.0183$.

(ii) $P(X = 8) = \dfrac{e^{-4}4^8}{8!} = \dfrac{65536\, e^{-4}}{40320} = 0.0298$.

(iii) $P(X \leq 2) = e^{-4}(1 + 4 + 8) = 13e^{-4} = 0.2381$.

(iv) $P(X = 4) = \dfrac{e^{-4}4^4}{4!} = 0.1954$.

14	A factory producing washing machines sends 1,000 machines each month to one of its customers. The manufacturer's records show that 5% of its machines are defective. The customer picks a sample of 20 machines and tests them. (i) What is the probability that the customer finds no defective machine in the sample? (ii) The customer tolerates at most 1 defective in the sample before accepting the consignment of 1,000 machines. What is the probability of accepting the whole consignment? (iii) What is the mean number of defective machines in the sample?	08	L3	CO2

The sample is only 2% of the lot, so the number of defectives is taken as binomial with $n = 20$, $p = 0.05$.

(i) $P(X = 0) = (0.95)^{20} = 0.3585$.

(ii) $P(X \leq 1) = (0.95)^{20} + 20(0.05)(0.95)^{19} = 0.3585 + 0.3774 = 0.7358$.

(iii) $E(X) = np = 20(0.05) = 1$ machine.

15	A multiple-choice test contains 25 questions, each with four answers. Assume a student just guesses on each question. (a) What is the probability that the student answers at least 20 questions correctly? (b) What is the probability that the student answers at most 5 questions correctly?	10	L3	CO2

The number correct is binomial with $n = 25$, $p = 0.25$.

(a) $P(X \geq 20) = \sum_{x=20}^{25} \binom{25}{x} (0.25)^x (0.75)^{25-x} = 1.24 \times 10^{-8}$, practically zero.

(b) $P(X \leq 5) = \sum_{x=0}^{5} \binom{25}{x} (0.25)^x (0.75)^{25-x} = 0.3783$.

16	The random variable X has a binomial distribution with $n = 10$ and $p = 0.01$. Determine (i) $P(X = 5)$ (ii) $P(X \leq 2)$ (iii) $P(X \geq 9)$ (iv) $P(3 \leq X < 5)$.	10	L3	CO2

$P(X = x) = \binom{10}{x}(0.01)^x(0.99)^{10-x}$.

(i) $P(X = 5) = 252(0.01)^5(0.99)^5 = 2.396 \times 10^{-8}$.

(ii) $P(X \leq 2) = (0.99)^{10} + 10(0.01)(0.99)^9 + 45(0.01)^2(0.99)^8 = 0.9044 + 0.0914 + 0.0042 = 0.99989$.

(iii) $P(X \geq 9) = 10(0.01)^9(0.99) + (0.01)^{10} = 9.91 \times 10^{-18} \approx 0$.

(iv) $P(3 \leq X < 5) = P(3) + P(4) = 120(0.01)^3(0.99)^7 + 210(0.01)^4(0.99)^6 = 1.118 \times 10^{-4} + 1.977 \times 10^{-6} = 1.138 \times 10^{-4}$.

17	Assume that each of your calls to a popular radio station has a probability of 0.02 of connecting (not getting a busy signal), and that your calls are independent. (a) What is the probability that your first call that connects is your tenth call? (b) What is the probability that more than five calls are needed to connect? (c) What is the mean number of calls needed to connect?	10	L3	CO2

The number of calls up to the first connection is geometric with $p = 0.02$.

(a) $P(X = 10) = (0.98)^9(0.02) = 0.0167$.

(b) $P(X > 5) =$ the first five calls all fail $= (0.98)^5 = 0.9039$.

(c) $E(X) = 1/p = 1/0.02 = 50$ calls.

18	What are the assumptions and applications of the binomial and Poisson distributions? Compare them.	08	L2	CO2

Binomial — assumptions: a fixed number n of trials; each trial has two outcomes; the success probability p is constant; the trials are independent. $P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$, mean np, variance $np(1-p)$. Applications: defectives in a sample of fixed size, acceptance sampling, the number of correct guesses in a test.

Poisson — assumptions: events occur one at a time and independently, at a constant average rate λ over a continuous interval of time, length, area or volume; the counts in disjoint intervals are independent. $P(X = x) = e^{-\lambda}\lambda^x / x!$, mean = variance = λ. Applications: calls per hour, flaws per square metre of cloth, accidents per day, arrivals at a counter.

Point	**Binomial**	**Poisson**
Count of	Successes in n trials	Events in an interval
Range	$0, 1, \ldots, n$	$0, 1, 2, \ldots$ (no upper limit)
Parameters	n and p	λ only
Mean and variance	np and $np(1-p)$; variance $<$ mean	Both equal λ

The Poisson is the limit of the binomial when $n \to \infty$ and $p \to 0$ with $np = \lambda$ fixed, so it approximates the binomial for large n and small p.

UNIT III – CONTINUOUS PROBABILITY DISTRIBUTIONS AND SAMPLING DISTRIBUTION

Continuous uniform, normal and exponential distributions; normal approximation; sampling distribution of the mean; Central Limit Theorem

Sl. No.	Questions	M	BT	CO
	PART – A (Short-answer questions)			
1	What is the shape of the graph of a continuous uniform distribution?	1	L1	CO2
	A rectangle: the density is a horizontal line of constant height $\frac{1}{b-a}$ over the interval $[a, b]$ and zero elsewhere.			
2	The area under the curve of a normal distribution equals _____.	1	L1	CO2
	One (the total probability).			
3	Enumerate any two real-life situations where the continuous uniform distribution applies.	2	L2	CO2
	(1) The waiting time for a bus or train that runs every 15 minutes when the passenger arrives at a random time: uniform on $[0, 15]$ min. (2) A dimension known only to lie within its tolerance band with every value equally likely, such as a flange thickness uniform between 0.95 and 1.05 mm; also the rounding error of a measurement.			
4	What is meant by the standard normal distribution?	2	L1	CO2
	The normal distribution with mean 0 and variance 1, $Z \sim N(0, 1)$. Any normal variable is converted to it by $Z = \dfrac{X - \mu}{\sigma}$, so one table of $\Phi(z) = P(Z \leq z)$ serves all normal distributions.			
5	What is the memoryless property of the exponential distribution?	2	L2	CO2
	For an exponential random variable X, $P(X > s + t \mid X > s) = P(X > t)$ for all $s, t \geq 0$. The probability of surviving a further time t does not depend on how long the item has already lasted; for example, the waiting time to the next Poisson arrival does not depend on how long one has already waited.			
6	In a customer service centre the time between arrivals is exponentially distributed with a mean of 10 minutes. Find the probability that the next customer arrives after 15 minutes.	2	L3	CO2
	$\lambda = 1/10$ per minute, so $P(X > 15) = e^{-15/10} = e^{-1.5} = 0.2231$.			
7	A fair coin is tossed 100 times. Using the normal approximation, find the probability of getting more than 60 heads.	2	L3	CO2
	$\mu = np = 50$, $\sigma = \sqrt{np(1-p)} = 5$. With the continuity correction, $P(X > 60) = P(X \geq 61) \approx P\big(Z > \frac{60.5-50}{5}\big) = P(Z > 2.1) = 0.0179$.			

8	For a normal distribution with $\mu = 70$ and $\sigma = 8$, find $P(62 < X < 78)$.	2	L3	CO2

$P(62 < X < 78) = P\left(\frac{62-70}{8} < Z < \frac{78-70}{8}\right) = P(-1 < Z < 1) = 0.8413 - 0.1587 = 0.6827.$

9	X is uniformly distributed on $[2, 6]$. Find $P(3 < X < 5)$.	2	L3	CO2

$f(x) = \frac{1}{6-2} = 0.25$, so $P(3 < X < 5) = (5 - 3)(0.25) = 0.5$.

10	The mean of the exponential distribution equals the mean of the related Poisson distribution only when the Poisson mean is equal to _____.	1	L2	CO2

One. If events occur at rate λ per unit time, the Poisson count per unit time has mean λ and the exponential time between events has mean $1/\lambda$; these are equal only when $\lambda = 1$.

11	A component has an average lifetime of 5 hours. Assuming an exponential distribution, find the probability that the component lasts more than 8 hours.	2	L3	CO2

$\lambda = 1/5$ per hour, so $P(X > 8) = e^{-8/5} = e^{-1.6} = 0.2019$.

12	The Central Limit Theorem states that, for large samples, the sampling distribution of the sample mean is approximately _____, whatever the population distribution.	1	L1	CO3

Normal, with mean μ and standard deviation $\sigma/\sqrt{n}$.

13	As the sample size increases, the standard error of the mean _____.	1	L2	CO3

Decreases, since $\sigma_{\bar{X}} = \sigma/\sqrt{n}$ (it halves when n is multiplied by 4).

14	The Central Limit Theorem is especially useful when the population distribution is _____.	1	L2	CO3

Unknown or not normal: the sample mean is still approximately normal for large n.

15	What is the difference between the population standard deviation and the standard error?	2	L2	CO3

The population standard deviation σ measures the spread of individual values in the population. The standard error $\sigma/\sqrt{n}$ is the standard deviation of the sample mean over repeated samples of size n; it measures how precisely $\bar{X}$ estimates μ and is smaller than σ for $n > 1$.

16	The mean weight of apples is 150 g with a standard deviation of 30 g. For a sample of 36 apples, find the probability that the sample mean is between 145 g and 155 g.	2	L3	CO3

$\sigma_{\bar{X}} = 30/\sqrt{36} = 5$ g, so $P(145 < \bar{X} < 155) = P(-1 < Z < 1) = 0.6827$.

17	Why is the sampling distribution important in statistics?	2	L2	CO3

A statistic such as $\bar{X}$ varies from sample to sample. Its sampling distribution tells how much it varies, which lets us attach probabilities to the statistic and so build confidence intervals and carry out hypothesis tests about population parameters from a single sample.

18	The standard deviation of a sampling distribution is also known as the _____.	1	L1	CO3

Standard error (for the sample mean, $\sigma/\sqrt{n}$).

19	If a population has mean μ and standard deviation σ, the standard error of the mean is $\sigma/\sqrt{n}$, where n is the _____.	1	L1	CO3

Sample size.

20	Define the sampling distribution of the sample mean.	2	L1	CO3

It is the probability distribution of $\bar{X}$ over all possible random samples of a fixed size n from a population. It has mean $\mu_{\bar{X}} = \mu$ and standard deviation $\sigma_{\bar{X}} = \sigma/\sqrt{n}$, and it is normal if the population is normal, or approximately normal for large n by the Central Limit Theorem.

Sl. No.	**Questions**	**M**	**BT**	**CO**
	PART – B (Descriptive and numerical questions)			
1	Describe the characteristics and applications of the normal distribution.	06	L2	CO2

Characteristics: density $f(x) = \dfrac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$, $-\infty < x < \infty$.

- Bell-shaped, unimodal and symmetric about μ; mean = median = mode = μ.
- Completely specified by μ (location) and σ (spread); the curve approaches the axis but never touches it.
- Total area 1; about 68.27%, 95.45% and 99.73% of the values lie within $\mu \pm \sigma$, $\mu \pm 2\sigma$, $\mu \pm 3\sigma$.
- Any normal variable is standardised by $Z = (X - \mu)/\sigma$; sums and means of independent normal variables are normal.

Applications: measurement errors and dimensions of manufactured parts; heights, weights and test scores; process capability and control charts; approximation of the binomial and Poisson for large counts; and, through the Central Limit Theorem, confidence intervals and tests on means.

2	Inter-arrival times (in minutes) between successive customers at a hotel are exponentially distributed with a mean of 2 minutes. (i) Compute the probability that the inter-arrival time between two customers is at most 2 minutes. (ii) What is the probability that at least 2 customers arrive in one minute? (iii) What is the probability that no customer arrives in a span of 3 minutes?	10	L3	CO2

Inter-arrival time $T \sim$ exponential with $\lambda = 1/2$ per minute; arrivals form a Poisson process with rate 0.5 per minute.

(i) $P(T \le 2) = 1 - e^{-2/2} = 1 - e^{-1} = 0.6321$.

(ii) In 1 min the number of arrivals is Poisson with mean 0.5: $P(N \ge 2) = 1 - e^{-0.5}(1 + 0.5) = 0.0902$.

(iii) In 3 min the mean is 1.5: $P(N = 0) = e^{-1.5} = 0.2231$ (equivalently $P(T > 3) = e^{-3/2}$).

3	A large conference room at a company can be reserved for no more than 4 hours. Both long and short meetings occur often; the length X of a meeting can be assumed equally likely anywhere in the interval $[1.5,\ 4]$ hours. (i) What is the probability density function? (ii) What is the probability that a given meeting lasts at least 3 hours? (iii) Find the mean and variance of the length X of the meetings.	10	L3	CO2

(i) $f(x) = \dfrac{1}{4-1.5} = 0.4$ for $1.5 \le x \le 4$, and 0 otherwise.
(ii) $P(X \ge 3) = (4-3)(0.4) = 0.4$.
(iii) $E(X) = \dfrac{1.5+4}{2} = 2.75$ h; $\quad V(X) = \dfrac{(4-1.5)^2}{12} = \dfrac{6.25}{12} = 0.5208$ h^2 (standard deviation 0.722 h).

4	A certain type of storage battery lasts, on average, 3.0 years with a standard deviation of 0.5 year. Assuming that battery life is normally distributed, find (i) the probability that a given battery will last less than 2.3 years; (ii) the probability that a battery lasts between 3.8 and 4.5 years; (iii) the life that 90% of the batteries exceed.	10	L3	CO2

(i) $P(X < 2.3) = P\big(Z < \frac{2.3-3}{0.5}\big) = P(Z < -1.4) = 0.0808$.
(ii) $P(3.8 < X < 4.5) = P(1.6 < Z < 3.0) = 0.9987 - 0.9452 = 0.0534$.
(iii) $P(X > x) = 0.90 \Rightarrow \frac{x-3}{0.5} = -1.2816$ (the z-value with 10% of the area below it), so $x = 3 - 1.2816(0.5) = 2.359$ years.

5	The response time X at an online computer terminal (the time between the end of a user's inquiry and the beginning of the system's response) has an exponential distribution with an expected response time of 5 s. Compute (i) $E(X)$; (ii) the probability that the response time is at most 10 s; (iii) the probability that the response time is between 5 and 10 s; (iv) give one more instance where the exponential distribution is frequently used.	10	L3	CO2

$\lambda = 1/5$ per second, $F(x) = 1 - e^{-x/5}$.
(i) $E(X) = 1/\lambda = 5$ s.
(ii) $P(X \le 10) = 1 - e^{-2} = 0.8647$.
(iii) $P(5 \le X \le 10) = e^{-1} - e^{-2} = 0.2325$.
(iv) The time between successive arrivals in a Poisson process (customers at a bank counter, calls at an exchange), or the time to failure of an electronic component with a constant failure rate.

6	The diameter of the dot produced by a printer is normally distributed with a mean of 0.002 inch and a standard deviation of 0.0004 inch. (i) What is the probability that the diameter of a dot exceeds 0.0026 inch? (ii) What is the probability that a diameter is between 0.0014 and 0.0028 inch? (iii) What standard deviation of diameters is needed so that the probability in part (ii) is 0.995?	10	L4	CO2

(i) $P(X > 0.0026) = P\big(Z > \frac{0.0026-0.002}{0.0004}\big) = P(Z > 1.5) = 0.0668$.
(ii) $P(0.0014 < X < 0.0028) = P(-1.5 < Z < 2) = 0.9772 - 0.0668 = 0.9104$.
(iii) The limits are 0.0006 below and 0.0008 above the mean, so we need $\Phi\big(\frac{0.0008}{\sigma}\big) - \Phi\big(-\frac{0.0006}{\sigma}\big) = 0.995$. Solving numerically gives $\sigma = 2.3122 \times 10^{-4}$ inch, about 0.00022 inch (check: z-limits -2.595 and 3.46 enclose 0.995).

7	An electrical firm manufactures light bulbs whose life before burn-out is normally distributed with a mean of 800 hours and a standard deviation of 40 hours. Find (i) the probability that a bulb burns between 778 and 834 hours; (ii) the probability that a bulb has a life of at least 1000 hours; (iii) the percentage of bulbs whose life is within $\pm 2\sigma$ of the mean life.	10	L3	CO2

(i) $P(778 < X < 834) = P(-0.55 < Z < 0.85) = 0.8023 - 0.2912 = 0.5112.$
(ii) $P(X \geq 1000) = P(Z \geq 5) = 2.87 \times 10^{-7} \approx 0.$
(iii) $P(\mu - 2\sigma < X < \mu + 2\sigma) = P(-2 < Z < 2) = 0.9545$, i.e. about 95.45% of bulbs (720 to 880 hours).

8	X is normally distributed with a mean of 10 and a standard deviation of 2. Determine (i) $P(X < 13)$ (ii) $P(X > 9)$ (iii) $P(6 < X < 14)$ (iv) $P(2 < X < 4)$.	10	L3	CO2

$Z = (X - 10)/2.$
(i) $P(Z < 1.5) = 0.9332.$
(ii) $P(Z > -0.5) = 0.6915.$
(iii) $P(-2 < Z < 2) = 0.9545.$
(iv) $P(-4 < Z < -3) = 0.00135 - 0.00003 = 0.00132.$

9	BBMP in Bangalore has installed 2000 electric lamps for street lighting in an area. The lamps have a mean burning life of 1000 hours with a standard deviation of 200 hours; assume a normal distribution. (i) What is the probability that a lamp fails in the first 700 burning hours? (ii) What is the probability that a lamp fails between 900 and 1300 burning hours? (iii) What is the probability that a lamp fails between 899 and 901 burning hours? (iv) After how many burning hours would we expect 10% of the lamps to be left?	10	L3	CO2

(i) $P(X < 700) = P(Z < -1.5) = 0.0668$, i.e. about $2000(0.0668) \approx 134$ lamps.
(ii) $P(900 < X < 1300) = P(-0.5 < Z < 1.5) = 0.9332 - 0.3085 = 0.6247$ (about 1249 lamps).
(iii) $P(899 < X < 901) = P(-0.005 < Z < 0.005) = 0.00399$ (about 8 lamps).
(iv) $P(X > x) = 0.10 \Rightarrow x = 1000 + 1.2816(200) = 1256.3$ hours.

10	The thickness of a conductive coating in micrometres has the density function $f(x) = 600x^{-2}$ for 100 μm $< x <$ 120 μm. (i) What is the mean coating thickness? (ii) What is the variance of the coating thickness? (iii) If the coating costs \$0.50 per micrometre of thickness on each part, what is the average cost of the coating per part?	10	L3	CO2

Check: $\int_{100}^{120} 600x^{-2}dx = 600\left(\frac{1}{100} - \frac{1}{120}\right) = 1.$

(i) $E(X) = \displaystyle\int_{100}^{120} x \cdot 600x^{-2}\,dx = 600 \ln \frac{120}{100} = 109.393\ \mu\text{m}.$

(ii) $E(X^2) = \displaystyle\int_{100}^{120} 600\,dx = 12000$, so $V(X) = 12000 - 109.393^2 = 33.19\ \mu\text{m}^2.$

(iii) Average cost $= 0.50\,E(X) = 0.50(109.393) = \$54.7$ per part.

11	The thickness of a flange on an aircraft component is uniformly distributed between 0.95 and 1.05 mm. Determine (a) the cumulative distribution function of flange thickness; (b) the proportion of flanges that exceed 1.02 mm; (c) the variance of flange thickness; (d) the thickness exceeded by 80% of the flanges.	10	L3	CO2

$f(x) = \frac{1}{0.10} = 10$ for $0.95 \le x \le 1.05$.

(a) $F(x) = 0$ for $x < 0.95$; $F(x) = 10(x - 0.95)$ for $0.95 \le x < 1.05$; $F(x) = 1$ for $x \ge 1.05$.

(b) $P(X > 1.02) = 1 - F(1.02) = 1 - 0.7 = 0.30$.

(c) $V(X) = \dfrac{(1.05 - 0.95)^2}{12} = 8.333 \times 10^{-4}$ mm^2.

(d) $P(X > x) = 0.80 \Rightarrow F(x) = 0.20 \Rightarrow x = 0.95 + 0.02 = 0.97$ mm.

12	The compressive strength of samples of cement can be modelled by a normal distribution with a mean of 6000 kg/cm^2 and a standard deviation of 100 kg/cm^2. (a) What is the probability that a sample's strength is less than 6250 kg/cm^2? (b) What is the probability that a sample's strength is between 5800 and 5900 kg/cm^2? (c) What strength is exceeded by 95% of the samples?	10	L3	CO2

(a) $P(X < 6250) = P(Z < 2.5) = 0.9938$.

(b) $P(5800 < X < 5900) = P(-2 < Z < -1) = 0.1587 - 0.0228 = 0.1359$.

(c) $P(X > x) = 0.95 \Rightarrow x = 6000 - 1.6449(100) = 5835.5$ kg/cm^2.

13	The time it takes a cell to divide (mitosis) is normally distributed with an average of one hour and a standard deviation of 5 minutes. (a) What is the probability that a cell divides in less than 45 minutes? (b) What is the probability that it takes a cell more than 65 minutes to divide? (c) What is the time by which approximately 99% of all cells complete mitosis?	10	L3	CO2

$\mu = 60$ min, $\sigma = 5$ min.

(a) $P(X < 45) = P(Z < -3) = 0.00135$.

(b) $P(X > 65) = P(Z > 1) = 0.1587$.

(c) $P(X < x) = 0.99 \Rightarrow x = 60 + 2.3263(5) = 71.63$ min.

14	The log-ons to a computer network follow a Poisson process with an average of five log-ons per minute. (a) What is the mean time between log-ons? (b) Determine the probability density and cumulative distribution functions of the time between log-ons. (c) Determine x such that the probability that at least one log-on occurs before time x minutes is 0.95. (d) What is the probability that more than 20 log-ons occur in 5 minutes?	10	L3	CO2

The time T between log-ons is exponential with $\lambda = 5$ per minute.

(a) $E(T) = 1/5 = 0.2$ min $= 12$ s.

(b) $f(t) = 5e^{-5t}$ and $F(t) = 1 - e^{-5t}$ for $t \ge 0$ (zero for $t < 0$).

(c) $P(T \le x) = 1 - e^{-5x} = 0.95 \Rightarrow x = \dfrac{\ln 20}{5} = 0.5991$ min (about 35.9 s).

(d) The count in 5 min is Poisson with mean 25: $P(N > 20) = 1 - \sum_{k=0}^{20} \dfrac{e^{-25}25^k}{k!} = 1 - 0.1855 = 0.8145$.

15	What are the key differences between a population and a sample? Why is sampling important in statistics?	06	L2	CO3

Point	**Population**	**Sample**
Meaning	The complete set of items or measurements of interest	A subset selected from the population
Size	Often very large or infinite (N)	Finite and manageable (n)
Measures	Parameters, fixed but usually unknown: μ, σ^2, p	Statistics, which vary between samples: $\bar{x}$, s^2, $\hat{p}$
Example	All bulbs made by a plant this year	50 bulbs tested for life

Why sampling: measuring the whole population is often impossible (infinite or future output), too costly or slow, or destructive (life tests, crash tests). A properly drawn random sample gives estimates of the parameters whose precision can be stated through the sampling distribution, so conclusions about the population can be drawn at a fraction of the cost.

16	A machine fills bottles with a mean volume of 500 ml and a standard deviation of 5 ml. For a random sample of 36 bottles, find the probability that the sample mean lies (i) between 498 and 502 ml; (ii) between 496 and 504 ml.	10	L3	CO3

By the Central Limit Theorem $\bar{X} \approx N\big(500,\ (5/\sqrt{36})^2\big)$, with standard error $5/6 = 0.8333$ ml.

(i) $P(498 < \bar{X} < 502) = P(-2.4 < Z < 2.4) = 0.9918 - 0.0082 = 0.9836.$

(ii) $P(496 < \bar{X} < 504) = P(-4.8 < Z < 4.8) = 0.999998 \approx 1.$

17	What is the Central Limit Theorem (CLT)? State the conditions for its applicability and its significance (any four applications) in statistical inference.	08	L2	CO3

Statement: if $X_1, \ldots, X_n$ is a random sample from a population with mean μ and finite variance σ^2, then for large n the sample mean is approximately normal:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \longrightarrow N(0,1) \quad \text{as } n \to \infty,$$

whatever the shape of the population.

Conditions: random sampling with independent observations; finite population mean and variance; a sufficiently large sample (usually $n \geq 30$; less if the population is nearly symmetric, more if it is strongly skewed).

Significance and applications:

- Normal-based probabilities for $\bar{X}$ even when the population is not normal.
- Confidence intervals for a population mean and for proportions.
- z-tests of hypotheses on means and proportions.
- Control charts for averages ($\bar{X}$ charts) in statistical quality control.
- Normal approximation to the binomial and Poisson distributions for large counts.

18	The life of a semiconductor laser at a constant power is normally distributed with a mean of 7000 hours and a standard deviation of 600 hours. (a) What is the probability that a laser fails before 6000 hours? (b) What is the life in hours that 95% of the lasers exceed? (c) If three lasers are used in a product and fail independently, what is the probability that all three are still operating after 7000 hours?	10	L3	CO2

(a) $P(X < 6000) = P\big(Z < \frac{6000-7000}{600}\big) = P(Z < -1.667) = 0.0478.$

(b) $P(X > x) = 0.95 \Rightarrow x = 7000 - 1.6449(600) = 6013.1$ hours.

(c) $P(X > 7000) = 0.5$ for each laser (7000 h is the mean), so by independence $P(\text{all three}) = (0.5)^3 = 0.125.$

UNIT IV – SIMPLE LINEAR REGRESSION AND CORRELATION

Empirical models; least-squares estimation; estimating σ^2; coefficient of determination; prediction; correlation; abuses of regression; reading Minitab output

Sl. No.	Questions	M	BT	CO
	PART – A (Short-answer questions)			
1	In a regression analysis, $S_{xy} = 385$ and the fitted relation is $\hat{y} = 2 + 5x$. Find the regression sum of squares.	2	L3	CO4
	$SS_R = \hat{\beta}_1 S_{xy} = 5(385) = 1925$.			
2	The relation $y = 2 + 3x_1 + 4x_2$ has how many independent variables?	1	L1	CO4
	Two (x_1 and x_2); it is a multiple linear regression model.			
3	Write the residual for the model $\hat{y} = a + bx$ corresponding to an observed value y_i.	1	L1	CO4
	$e_i = y_i - \hat{y}_i = y_i - (a + bx_i)$, the observed value minus the fitted value.			
4	From Minitab output, the intercept is 2.5 and the slope is 1.8. If $X = 6$, find the predicted value of Y.	2	L3	CO4
	$\hat{Y} = 2.5 + 1.8(6) = 2.5 + 10.8 = 13.3$.			
5	Given X: 1, 2, 3, 4 and Y: 2, 4, 5, 4, find the least-squares estimates of β_0 and β_1.	2	L3	CO4
	$\bar{x} = 2.5$, $\bar{y} = 3.75$, $S_{xy} = \sum xy - n\bar{x}\bar{y} = 41 - 37.5 = 3.5$, $S_{xx} = 30 - 25 = 5$. $\hat{\beta}_1 = 3.5/5 = 0.7$ and $\hat{\beta}_0 = 3.75 - 0.7(2.5) = 2$, so $\hat{y} = 2 + 0.7x$.			
6	The method of estimating β_0 and β_1 by minimising the sum of squared errors is called the _____ method.	1	L1	CO4
	Method of least squares.			
7	In simple linear regression, R^2 gives the proportion of the variation in _____ explained by the model.	1	L1	CO4
	The response (dependent) variable Y.			
8	Predicting far outside the range of the observed data is called _____.	1	L1	CO4
	Extrapolation; it is risky because the linear relation may not hold outside the range of the data.			

9	A common abuse of regression is using it with _____ variables without checking for linearity.	1	L2	CO4

Categorical (non-quantitative) variables, or variables whose relation is not linear; a scatter plot should be checked first.

10	State two assumptions of the simple linear regression model.	2	L1	CO4

(1) The mean of Y is a linear function of x: $E(Y \mid x) = \beta_0 + \beta_1 x$. (2) The errors ϵ are independent and normally distributed with mean 0 and constant variance σ^2.

11	What does a correlation coefficient of -0.85 indicate?	2	L2	CO4

A strong negative linear relationship: as one variable increases, the other tends to decrease, and the points lie fairly close to a downward-sloping line ($r^2 = 0.7225$, so about 72% of the variation is explained by the linear relation).

12	List one Minitab output used to check the significance of a regression model.	2	L1	CO4

The P-value of the slope coefficient (its t-statistic) or the P-value of the F-test in the analysis-of-variance table; a value below α (say 0.05) shows the regression is significant. R-Sq is also reported.

13	The strength of a regression relation is measured by _____.	1	L1	CO4

The coefficient of determination R^2 (or the correlation coefficient r).

14	In a regression analysis, $S_{xy} = 385$, $S_{yy} = 2308.3$ and $\hat{y} = 2 + 5x$. Find R^2.	2	L3	CO4

$SS_R = \hat{\beta}_1 S_{xy} = 1925$, so $R^2 = \dfrac{SS_R}{S_{yy}} = \dfrac{1925}{2308.3} = 0.8339$.

15	For a regression, $S_{xx} = 53$, $S_{xy} = 48$ and $S_{yy} = 84$. Find SS_E.	2	L3	CO4

$SS_E = S_{yy} - \dfrac{S_{xy}^2}{S_{xx}} = 84 - \dfrac{48^2}{53} = 84 - 43.472 = 40.528$.

16	When a set of points is fitted with a second-degree curve, what type of regression is performed?	1	L1	CO4

Polynomial (quadratic) regression, $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon$.

17	In linear regression, the error term follows which distribution (with its parameters)?	2	L1	CO4

A normal distribution with mean 0 and constant variance: $\epsilon \sim N(0, \sigma^2)$, independently for each observation.

18	Define the correlation coefficient and state its range.	2	L1	CO4

$r = \dfrac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$ measures the strength and direction of the linear relationship between two variables. $-1 \le r \le 1$: $r = \pm 1$ is a perfect linear relation, $r = 0$ means no linear relation, and the sign gives the direction.

19	When can you be confident that a regression relation is proper and usable for prediction?	2	L2	CO4

When the slope is significant (small P-value), R^2 is high, the residuals show no pattern and look normal with constant variance, and predictions are made within the range of the data used to fit the model.

20	State the assumptions of the linear regression model.	2	L1	CO4

Linearity of the mean response in x; independence of the errors; constant error variance (homoscedasticity); normally distributed errors with mean zero; x measured without error.

Sl. No.	**Questions**	**M**	**BT**	**CO**
	PART – B (Descriptive and numerical questions)			
1	What are the assumptions of simple linear regression? How do you validate them using residual plots?	06	L2	CO4

Model: $Y = \beta_0 + \beta_1 x + \epsilon$ with (1) linearity of $E(Y \mid x)$ in x; (2) independent errors; (3) constant error variance σ^2; (4) normally distributed errors with mean 0.
Validation with residuals $e_i = y_i - \hat{y}_i$:

- Residuals against fitted values (or x): a structureless horizontal band about zero supports linearity and constant variance; a curve indicates non-linearity, and a funnel indicates non-constant variance.
- Normal probability plot (or histogram) of residuals: points close to a straight line support normality; outliers appear as isolated points (standardised residuals beyond ± 3).
- Residuals in time order: no runs or trends supports independence.

2	Describe the method of least squares in simple linear regression. How is the best-fit line determined?	06	L2	CO4

For data (x_i, y_i), $i = 1, \ldots, n$, the least-squares line is the one that minimises the sum of squared vertical deviations

$$L = \sum_{i=1}^{n} \big(y_i - \beta_0 - \beta_1 x_i\big)^2.$$

Setting $\partial L/\partial \beta_0 = 0$ and $\partial L/\partial \beta_1 = 0$ gives the normal equations, whose solution is

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x},$$

with $S_{xx} = \sum x_i^2 - \frac{(\sum x_i)^2}{n}$ and $S_{xy} = \sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n}$. The fitted line $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ passes through $(\bar{x}, \bar{y})$, its residuals add to zero, and $\hat{\sigma}^2 = SS_E/(n-2)$ estimates the error variance.

3	An online retailer wants to predict customer spending from the number of website visits. Explain how linear regression applies in this situation and discuss its advantages and limitations.	06	L3	CO4

Let x = number of website visits (predictor) and Y = spending (response). From historical customer data: plot Y against x to check that the relation is roughly linear; fit $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ by least squares; interpret $\hat{\beta}_1$ as the average extra spending per additional visit; check R^2, the significance of the slope and the residual plots; then predict spending for given visit levels within the observed range.
Advantages: simple to fit and explain; quantifies the effect of visits on spending; gives predictions with an error estimate; supports targeting and budgeting decisions.
Limitations: assumes a straight-line relation and constant variance; sensitive to outliers (a few big spenders); shows association, not causation; ignores other drivers such as income or promotions; unreliable when extrapolated beyond the observed number of visits.

4	What are the ways in which linear regression techniques are misused?	06	L2	CO4

- **Extrapolation:** predicting well outside the range of the data, where the relation may be different.
- **Inferring causation:** a strong association does not prove that x causes Y; both may be driven by a third variable.
- **Ignoring the scatter plot:** fitting a straight line to a curved relation, or to data dominated by one or two outliers or influential points.
- **Ignoring the assumptions:** not checking residuals for non-constant variance, non-normality or dependence, so that tests and intervals are invalid.
- **Judging fit by R^2 alone,** or using regression with categorical, non-quantitative data, or with too few observations.

5	Consider a simple linear regression model $y = \beta_0 + \beta_1 x + \epsilon$ based on n observations. Summarised data on 10 observations give $\sum(y_i - \hat{y}_i)^2 = 3.267$, $\sum y_i = 63.3$, $\sum y_i^2 = 423.49$, $\sum x_i = 139$ and $\sum x_i^2 = 2239$. (i) Estimate β_1 and β_0, assuming a positive correlation between X and Y. (ii) Obtain an estimate of the underlying variance. (iii) What proportion of the variability is explained by the regression model? (iv) Assuming normally distributed errors, how will a plot of residuals against predicted values appear? (Show an indicative sketch.)	10	L4	CO4

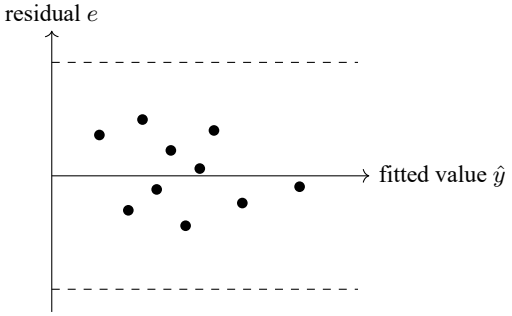
$S_{xx} = 2239 - \frac{139^2}{10} = 306.9$, $S_{yy} = 423.49 - \frac{63.3^2}{10} = 22.801$, and $SS_R = S_{yy} - SS_E = 22.801 - 3.267 = 19.534$.

(i) Since $SS_R = \hat{\beta}_1 S_{xy} = \hat{\beta}_1^2 S_{xx}$, $\hat{\beta}_1^2 = \dfrac{19.534}{306.9} = 0.06365$, and with positive correlation $\hat{\beta}_1 = +0.2523$. $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 6.33 - 0.2523(13.9) = 2.8232$, so $\hat{y} = 2.823 + 0.2523x$.

(ii) $\hat{\sigma}^2 = \dfrac{SS_E}{n-2} = \dfrac{3.267}{8} = 0.4084$.

(iii) $R^2 = \dfrac{SS_R}{S_{yy}} = \dfrac{19.534}{22.801} = 0.8567$: about 85.7% of the variability in Y is explained.

(iv) The residuals should form a random, patternless horizontal band of roughly constant width centred on zero, with nearly all points within about $\pm 2\hat{\sigma} = \pm 1.28$:



6	In an oil refinery, chemical reactions are observed for a set of 20 pairs of values (x_i, y_i), giving $\sum x_i^2 = 130000$, $\sum y_i^2 = 88.6$, $\sum x_i y_i = 1500$, $\sum x_i = 1500$ and $\sum y_i = 10$. (i) Compute the intercept and slope and write the estimated regression model relating x and y. (ii) Compute the predicted value of y for $x = 85$ and the corresponding residual if the observed value is 0.8. (iii) Estimate the error variance and the proportion of variability not explained by the model.	10	L3	CO4

$S_{xx} = 130000 - \frac{1500^2}{20} = 17500$, $S_{xy} = 1500 - \frac{1500(10)}{20} = 750$, $S_{yy} = 88.6 - \frac{10^2}{20} = 83.6$; $\bar{x} = 75$, $\bar{y} = 0.5$.

(i) $\hat{\beta}_1 = \frac{750}{17500} = 0.04286$, $\hat{\beta}_0 = 0.5 - 0.04286(75) = -2.7143$, so $\hat{y} = -2.7143 + 0.04286x$.

(ii) $\hat{y}(85) = -2.7143 + 0.04286(85) = 0.9286$; residual $= 0.8 - 0.9286 = -0.1286$.

(iii) $SS_R = \hat{\beta}_1 S_{xy} = 32.143$, $SS_E = 83.6 - 32.143 = 51.457$, $\hat{\sigma}^2 = \frac{51.457}{18} = 2.8587$. $R^2 = 0.3845$, so the proportion not explained is $1 - R^2 = 0.6155$ (about 61.6%).

7	During the fall harvest season pumpkins are sold in large quantities. Instead of weighing the pumpkins, the operator often places a pumpkin in the appropriate circular cut-out on the counter; one farmer said, "I can tell the weight of the pumpkin from its circumference." To check this, 23 pumpkins were measured for circumference and weight:	16	L4	CO4

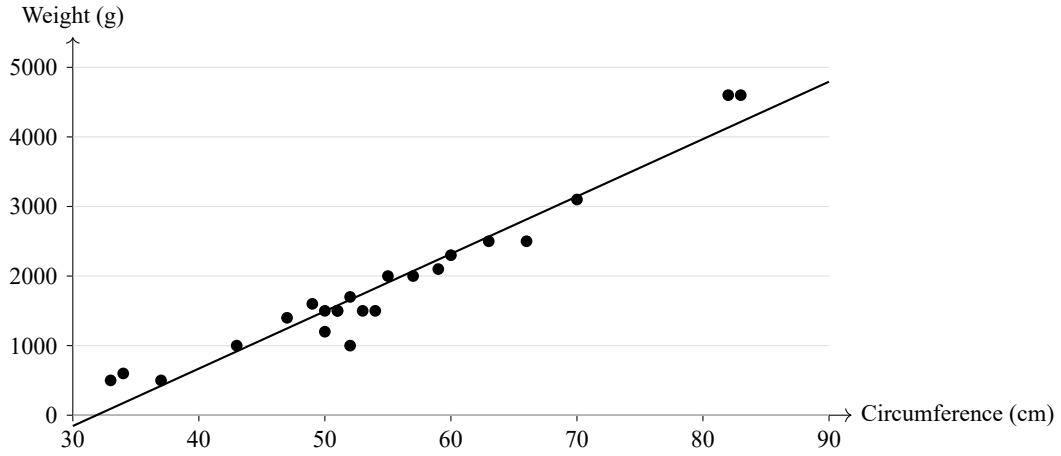
Circ. (cm)	Weight (g)	Circ. (cm)	Weight (g)	Circ. (cm)	Weight (g)
50	1200	57	2000	47	1400
55	2000	66	2500	51	1500
54	1500	82	4600	63	2500
52	1700	83	4600	33	500
37	500	70	3100	43	1000
52	1000	34	600	50	1500
53	1500	51	1500	49	1600
60	2300	59	2100		

a. Assuming a linear relationship, use the least-squares method to compute the regression coefficients b_0 and b_1.
b. Interpret the meaning of the slope b_1.
c. Predict the mean weight of a pumpkin that is 60 cm in circumference.
d. Do you think it is a good idea for the farmer to sell pumpkins by circumference instead of weight? Explain.

$n = 23$, $\sum x = 1251$, $\sum y = 42700$, $\sum x^2 = 71601$, $\sum xy = 2615900$, $\sum y^2 = 105090000$.

$$S_{xx} = 3557.478, \quad S_{xy} = 293391.3, \quad S_{yy} = 25816521.74, \quad \bar{x} = 54.3913, \quad \bar{y} = 1856.522$$

a. $b_1 = \dfrac{S_{xy}}{S_{xx}} = 82.472$ g/cm and $b_0 = \bar{y} - b_1\bar{x} = -2629.22$ g, so $\hat{y} = -2629.22 + 82.472x$.

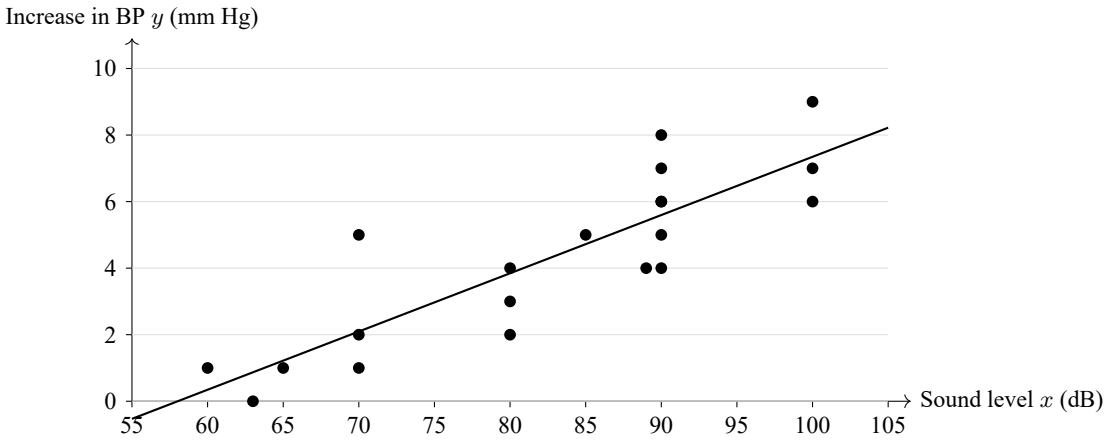


b. Each additional centimetre of circumference increases the mean weight by about 82.5 g. (The intercept has no physical meaning, since a 0 cm pumpkin is far outside the data.)
c. $\hat{y}(60) = -2629.22 + 82.472(60) = 2319.1$ g.
d. $r = 0.9681$ and $R^2 = 0.9372$: about 93.7% of the variation in weight is explained by circumference, with $\hat{\sigma} = 278$ g. Selling by circumference is a reasonable, quick approximation for typical pumpkins, but individual weights can differ by several hundred grams and the scatter suggests a slightly curved relation (weight grows faster than linearly with size), so for accurate pricing weighing is better.

8	Data on the increase in blood pressure y (mm Hg) against sound level exposure x (dB) for 20 instances are: x: 60, 63, 65, 70, 70, 70, 80, 90, 80, 80 y: 1, 0, 1, 2, 5, 1, 4, 6, 2, 3 x: 85, 89, 90, 90, 90, 90, 90, 100, 100, 100 y: 5, 4, 6, 8, 4, 5, 7, 9, 7, 6 (i) Draw a scatter diagram of y against x. Does a simple linear regression model seem reasonable? (ii) Fit the simple linear regression model using least squares and find an estimate of σ^2. (iii) Find the predicted blood pressure at a sound level of 85 dB. What is the residual in this case? (iv) Find the predicted blood pressure at sound levels of 75 dB and 105 dB.	16	L4	CO4

$n = 20$, $\sum x = 1652$, $\sum y = 86$, $\sum x^2 = 139440$, $\sum y^2 = 494$, $\sum xy = 7626$.

(i) Scatter diagram (with the fitted line from part ii):



The points rise steadily with x with roughly even scatter, so a straight-line model is reasonable.

(ii) $S_{xx} = 2984.8$, $S_{xy} = 522.4$, $S_{yy} = 124.2$. $\hat{\beta}_1 = 0.17502$, $\hat{\beta}_0 = 4.3 - 0.17502(82.6) = -10.1567$, so $\hat{y} = -10.1567 + 0.17502x$. $SS_E = S_{yy} - \hat{\beta}_1 S_{xy} = 32.769$ and $\hat{\sigma}^2 = \frac{SS_E}{18} = 1.8205$.

(iii) $\hat{y}(85) = 4.72$ mm Hg. The observed value at 85 dB is 5, so the residual is $5 - 4.72 = 0.28$.

(iv) $\hat{y}(75) = 2.97$ mm Hg; $\hat{y}(105) = 8.22$ mm Hg (105 dB lies outside the data, so this is an extrapolation and less reliable).

9	For a set of 20 pairs of values (x_i, y_i), $\sum x_i^2 = 29.29$, $\sum y_i^2 = 170044.53$, $\sum x_i y_i = 2214.66$, $\sum x_i = 23.92$ and $\sum y_i = 1843.21$. Write the regression line equation. It was later found that the pair $(x_{20}, y_{20}) = (2, 90)$ was wrong; it was deleted and the correct pair $(0, 60)$ inserted. Write the new regression line equation. What is the residual at $x = 0$ with the corrected data set?	10	L4	CO4

Original data: $S_{xx} = 29.29 - \frac{23.92^2}{20} = 0.6817$, $S_{xy} = 2214.66 - \frac{23.92(1843.21)}{20} = 10.1808$; $\hat{\beta}_1 = 14.9349$, $\hat{\beta}_0 = 92.1605 - 14.9349(1.196) = 74.298$, so $\hat{y} = 74.298 + 14.935x$.

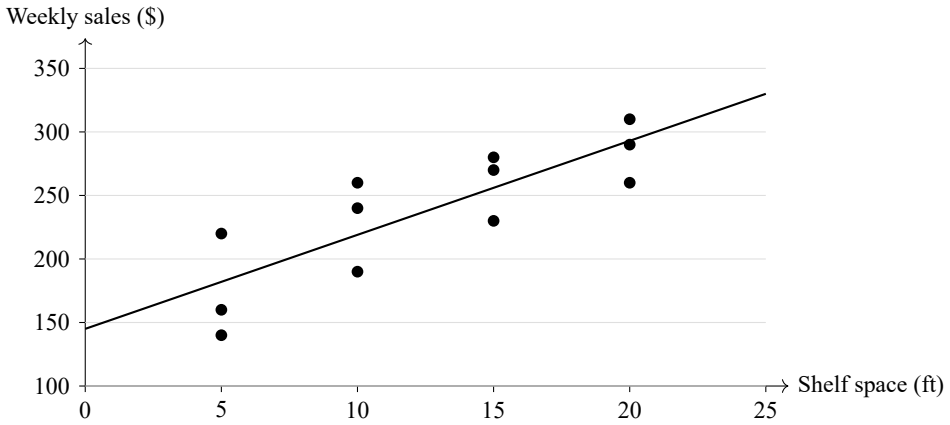
Corrected data: $\sum x = 21.92$, $\sum y = 1813.21$, $\sum x^2 = 29.29 - 4 + 0 = 25.29$, $\sum xy = 2214.66 - 180 + 0 = 2034.66$ ($\sum y^2 = 165544.53$). $S_{xx} = 1.2657$, $S_{xy} = 47.3818$; $\hat{\beta}_1 = 37.4359$, $\hat{\beta}_0 = 90.6605 - 37.4359(1.096) = 49.631$, so $\hat{y} = 49.631 + 37.436x$.

Residual at the corrected point $(0, 60)$: $e = 60 - 49.631 = 10.369$.

10	A store manager would like to use shelf space to predict the sales of a certain food item. A random sample of 12 equal-sized stores gives: a. Construct a scatter plot. b. Calculate the slope and intercept. c. Interpret the meaning of the slope b_1. d. Predict the mean weekly sales of the food item for stores with 8 feet of shelf space.	16	L3	CO4

Store	1	2	3	4	5	6	7	8	9	10	11	12
Space X (ft)	5	5	5	10	10	10	15	15	15	20	20	20
Sales Y ($)	160	220	140	190	240	260	230	270	280	260	290	310

a. Scatter plot (with the fitted line from part b):



b. $n = 12$, $\sum x = 150$, $\sum y = 2850$, $\sum x^2 = 2250$, $\sum xy = 38400$; $S_{xx} = 375$, $S_{xy} = 2775$. $b_1 = \frac{2775}{375} = 7.4$ and $b_0 = 237.5 - 7.4(12.5) = 145$, so $\hat{y} = 145 + 7.4x$.

c. Each additional foot of shelf space increases mean weekly sales by about \$7.4.

d. $\hat{y}(8) = 145 + 7.4(8) = \$204.2$ per week (\$2.042 hundred).

11	A physicist wishes to use the model $y = k/x^r$, where x is the independent and y the dependent variable, with the following data. Use the transformation $\ln y = \ln k - r \ln x$ and the least-squares estimators of linear regression. (i) Obtain estimates of k and r. (ii) What is the predicted value of y at $x = 60$, and the corresponding residual? (iii) Provide an estimate of the underlying variance of the y_i values.	16	L4	CO4

x_i	50	60	70	80	90
y_i	64.5	51.0	40.6	33.4	26.0

Let $u = \ln x$ and $v = \ln y$, so $v = \ln k - r\,u$ is a straight line with slope $-r$:

$u = \ln x$	3.912	4.0943	4.2485	4.382	4.4998
$v = \ln y$	4.1667	3.9318	3.7038	3.5086	3.2581

$\bar{u} = 4.2273$, $\bar{v} = 3.7138$, $S_{uu} = 0.21573$, $S_{uv} = -0.32792$, $S_{vv} = 0.50251$.

(i) Slope $= S_{uv}/S_{uu} = -1.5201$, so $\hat{r} = 1.5201$. Intercept $\ln \hat{k} = \bar{v} - (-1.5201)\bar{u} = 10.1396$, so $\hat{k} = e^{10.1396} = 25326$. Fitted model: $\hat{y} = 25326\,x^{-1.5201}$.

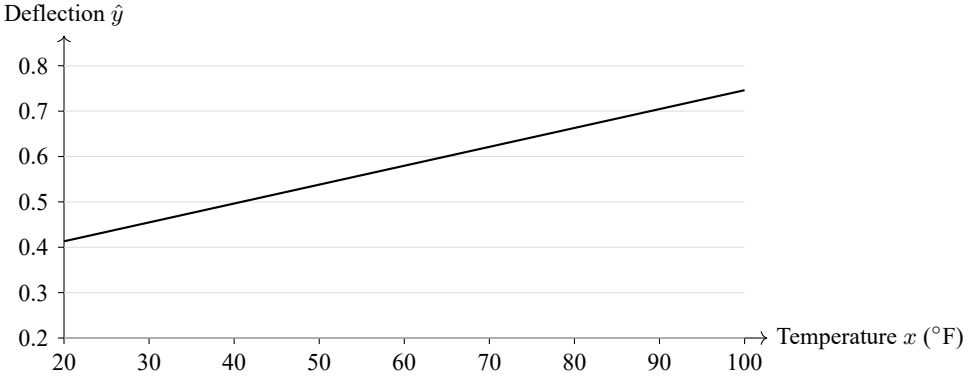
(ii) $\hat{y}(60) = 25326(60)^{-1.5201} = 50.196$; residual $= 51.0 - 50.196 = 0.804$.

(iii) In the fitted (log) scale, $\hat{\sigma}^2_{\ln y} = \dfrac{SS_E}{n-2} = \dfrac{0.004058}{3} = 1.353 \times 10^{-3}$ ($R^2 = 0.9919$). In the original units, $\sum(y_i - \hat{y}_i)^2 = 6.601$, so $\hat{\sigma}^2_y \approx \frac{6.601}{3} = 2.2$.

12	Regression methods were used to analyse data from a study of the relationship between roadway surface temperature x (°F) and pavement deflection y. Summary quantities: $n = 20$, $\sum y_i = 12.75$, $\sum y_i^2 = 8.86$, $\sum x_i = 1478$, $\sum x_i^2 = 143215.8$, $\sum x_i y_i = 1083.67$. a. Calculate the least-squares estimates of the slope and intercept, graph the regression line, and estimate σ^2. b. Use the fitted line to predict the pavement deflection when the surface temperature is 85°F. c. What is the mean pavement deflection when the surface temperature is 90°F? d. What change in mean pavement deflection is expected for a 1°F change in surface temperature?	10	L3	CO4

$S_{xx} = 143215.8 - \frac{1478^2}{20} = 33991.6$, $S_{xy} = 1083.67 - \frac{1478(12.75)}{20} = 141.445$, $S_{yy} = 8.86 - \frac{12.75^2}{20} = 0.73187$; $\bar{x} = 73.9$, $\bar{y} = 0.6375$.

a. $\hat{\beta}_1 = 0.004161$, $\hat{\beta}_0 = 0.6375 - 0.004161(73.9) = 0.33$, so $\hat{y} = 0.33 + 0.00416x$. $SS_E = 0.73187 - 0.004161(141.445) = 0.1433$ and $\hat{\sigma}^2 = SS_E/18 = 0.00796$.



b. $\hat{y}(85) = 0.6837$.

c. $\hat{\mu}_{Y|90} = 0.7045$.

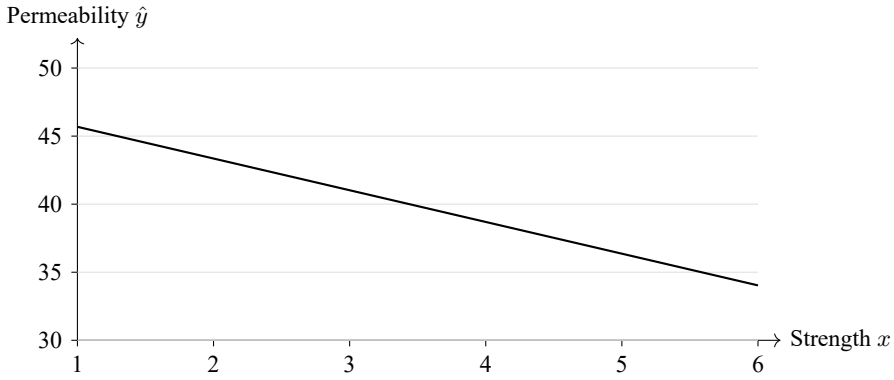
d. $\hat{\beta}_1 = 0.00416$ per °F.

13	An article in *Concrete Research* presented data on the compressive strength x and intrinsic permeability y of various concrete mixes and cures. Summary quantities: $n = 14$, $\sum y_i = 572$, $\sum y_i^2 = 23530$, $\sum x_i = 43$, $\sum x_i^2 = 157.42$, $\sum x_i y_i = 1697.80$. Assume the two variables are related by a simple linear regression model. a. Calculate the least-squares estimates of the slope and intercept, estimate σ^2, and graph the regression line. b. Use the fitted line to predict the permeability when the compressive strength is $x = 4.3$. c. Give a point estimate of the mean permeability when the compressive strength is $x = 3.7$. d. The observed permeability at $x = 3.7$ is $y = 46.1$. Calculate the corresponding residual.	10	L3	CO4

$S_{xx} = 157.42 - \frac{43^2}{14} = 25.3486$, $S_{xy} = 1697.80 - \frac{43(572)}{14} = -59.0571$, $S_{yy} = 23530 - \frac{572^2}{14} = 159.7143$; $\bar{x} = 3.0714$, $\bar{y} = 40.8571$.

a. $\hat{\beta}_1 = -2.3298$, $\hat{\beta}_0 = 40.8571 - (-2.3298)(3.0714) = 48.013$, so $\hat{y} = 48.013 - 2.3298x$. $SS_E = 159.7143 -$

$(-2.3298)(-59.0571) = 22.1229$, $\hat{\sigma}^2 = SS_E/12 = 1.8436$.



b. $\hat{y}(4.3) = 37.995$.
c. $\hat{\mu}_{Y|3.7} = 39.393$.
d. $e = 46.1 - 39.393 = 6.707$.

UNIT V – STATISTICAL INFERENCE AND INTERVAL ESTIMATION

Hypothesis testing; inference on the mean of a normal population (variance known and unknown) and on the variance; goodness of fit; confidence intervals on the mean and variance

Sl. No.	Questions	M	BT	CO
	PART – A (Short-answer questions)			
1	The null hypothesis is rejected when the P-value is _____ the level of significance.	1	L1	CO4
	Less than (or equal to) α.			
2	Differentiate between Type I and Type II errors in hypothesis testing.	2	L2	CO4
	A Type I error is rejecting H_0 when it is true; its probability is α, the significance level. A Type II error is failing to reject H_0 when it is false; its probability is β, and $1-\beta$ is the power of the test.			
3	Type I error is committed when a _____ hypothesis is _____.	2	L1	CO4
	A true (null) hypothesis is rejected.			
4	The test statistic used to test a population variance follows which distribution?	1	L1	CO4
	The chi-square distribution with $n-1$ degrees of freedom (for a normal population).			
5	The test statistic for a test on the variance of a normal population with sample size n is _____. State the meaning of each term.	2	L1	CO4
	$$\chi_0^2 = \frac{(n-1)S^2}{\sigma_0^2},$$ where n is the sample size, S^2 the sample variance and σ_0^2 the variance stated in H_0; under H_0 it has a χ^2 distribution with $n-1$ degrees of freedom.			
6	Write the null and alternative hypotheses to test whether a new process has reduced the population mean.	2	L2	CO4
	$H_0 : \mu = \mu_0$ (no reduction) against $H_1 : \mu < \mu_0$, a lower-tailed test, where μ_0 is the mean of the old process.			
7	A sample is collected to test whether the population variance exceeds 5. State the hypotheses.	2	L2	CO4
	$H_0 : \sigma^2 = 5$ against $H_1 : \sigma^2 > 5$ (upper-tailed).			
8	What is meant by the level of significance?	1	L1	CO4
	The probability of a Type I error that the analyst is willing to accept, $\alpha = P(\text{reject } H_0 \mid H_0 \text{ true})$, commonly 0.05 or 0.01; it fixes the critical region.			

9	When is a z-test used for inference on a population mean?	2	L1	CO4
When the population standard deviation σ is known and the population is normal, or when the sample is large ($n \geq 30$), so that $\bar{X}$ is approximately normal by the Central Limit Theorem.				

10	If the confidence level is increased from 95% to 99%, the width of the confidence interval _____.	1	L2	CO3
Increases (the critical value grows from 1.96 to 2.576 for a z-interval).				

11	The sample mean is 52, the population standard deviation 6 and the sample size 36. Construct a 95% confidence interval for the population mean.	2	L3	CO3
$\bar{x} \pm z_{0.025}\frac{\sigma}{\sqrt{n}} = 52 \pm 1.96\frac{6}{6} = 52 \pm 1.96$, i.e. $50.04 \leq \mu \leq 53.96$.				

12	The fill amount of bottles is $N(500, 16^2)$ ml. A sample of 25 bottles has a mean of 505 ml. Construct a 95% confidence interval for the mean.	2	L3	CO3
$505 \pm 1.96\frac{16}{5} = 505 \pm 6.272$, i.e. $498.73 \leq \mu \leq 511.27$ ml.				

13	A sample of 10 students has a mean of 75 and a sample standard deviation of 5. Construct a 95% confidence interval for the mean.	2	L3	CO3
σ is unknown, so use $t_{0.025,9} = 2.262$: $75 \pm 2.262\frac{5}{\sqrt{10}} = 75 \pm 3.577$, i.e. $71.42 \leq \mu \leq 78.58$.				

14	A sample of 15 items has a variance of 20. Test whether the population variance is 16, at the 5% significance level.	2	L3	CO4
$H_0 : \sigma^2 = 16$, $H_1 : \sigma^2 \neq 16$. $\chi_0^2 = \frac{14(20)}{16} = 17.5$. Critical values: $\chi^2_{0.975,14} = 5.629$ and $\chi^2_{0.025,14} = 26.119$. Since 17.5 lies between them, H_0 is not rejected.				

15	Find the 90th percentile of a χ^2 random variable with 9 degrees of freedom.	2	L3	CO3
The value with 90% of the probability below it: $\chi^2_{0.10,9} = 14.684$.				

Sl. No.	Questions	M	BT	CO
	PART – B (Descriptive and numerical questions)			
1	The length L of certain specialty parts is normally distributed with a variance of 25 mm^2. A customer specifies the length as 100 ± 2.00 mm and the decision is based on the mean of a sample of $n = 50$ parts. What is the probability of a Type I error in a test of $H_0 : \mu = 100$ mm against $H_1 : \mu \neq 100$ mm if H_0 is rejected when $\bar{x}$ falls outside 100 ± 2 mm?	10	L3	CO4

$\sigma_{\bar{X}} = \frac{5}{\sqrt{50}} = 0.7071$ mm. A Type I error occurs if $\bar{X} < 98$ or $\bar{X} > 102$ while $\mu = 100$:

$$\alpha = P\Big(|Z| > \frac{2}{0.7071}\Big) = 2P(Z > 2.828) = 2(0.00234) = 0.0047.$$

Answer: $\alpha = 0.0047$ (about 0.47%).

2	Explain the steps involved in conducting a hypothesis test.	10	L2	CO4

1. **Parameter:** identify the parameter of interest (for example the mean μ).
2. **Null hypothesis:** state H_0, e.g. $\mu = \mu_0$.
3. **Alternative:** state H_1 ($\mu \neq \mu_0$, $\mu > \mu_0$ or $\mu < \mu_0$), which decides one- or two-tailed testing.
4. **Significance level:** choose α (0.05 or 0.01).
5. **Test statistic:** choose it from the situation, e.g. $z_0 = \frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}}$ (σ known), $t_0 = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}$ (σ unknown), $\chi_0^2 = \frac{(n-1)s^2}{\sigma_0^2}$ (variance).
6. **Rejection region:** the critical values for α (or decide to compare the P-value with α).
7. **Compute** the statistic from the sample.
8. **Decide and conclude** in the context of the problem.

Example. The burning rate of a propellant should be 50 cm/s with $\sigma = 2$. A sample of $n = 25$ gives $\bar{x} = 51.3$. $H_0 : \mu = 50$, $H_1 : \mu \neq 50$, $\alpha = 0.05$; $z_0 = \frac{51.3-50}{2/5} = 3.25 > 1.96$ and $P = 2(1 - \Phi(3.25)) = 0.0012$, so H_0 is rejected: the mean burning rate differs from 50 cm/s.

3	(i) What are the hypotheses and the test statistic for a two-sided test on the variance of a population? What assumptions are needed, and what is the rejection region? (ii) State the probability that a t random variable with 25 degrees of freedom takes a value between 2.06 and 3.45. (iii) What is the value x such that a χ^2 random variable with 15 degrees of freedom is at most x with probability 0.95? (iv) In a factory, 50 lots are inspected each day and the number of defectives x_i in each lot is recorded, $i = 1, 2, \ldots, 50$. Each lot has 100 parts. The probability of a part being defective can be estimated as _____.	10	L3	CO4

(i) $H_0 : \sigma^2 = \sigma_0^2$ against $H_1 : \sigma^2 \neq \sigma_0^2$. Statistic $\chi_0^2 = \dfrac{(n-1)S^2}{\sigma_0^2}$, which has a χ^2_{n-1} distribution under H_0. Assumption: a random sample from a *normal* population (the test is sensitive to non-normality). Reject H_0 if $\chi_0^2 > \chi^2_{\alpha/2,n-1}$ or $\chi_0^2 < \chi^2_{1-\alpha/2,n-1}$.

(ii) $P(2.06 < T_{25} < 3.45) = 0.999 - 0.975 = 0.024$ (from tables, $t_{0.025,25} = 2.060$ and $t_{0.001,25} = 3.450$, giving $0.025 - 0.001 = 0.024$).

(iii) $x = \chi^2_{0.05,15} = 24.996$.

(iv) $\hat{p} = \dfrac{\sum_{i=1}^{50} x_i}{50 \times 100} = \dfrac{\sum x_i}{5000}$, the total number of defectives divided by the total number of parts inspected.

4	In the maintenance department of a firm, the life of an electronic sensor is found to be normally distributed. A random sample of 30 sensors has a mean life of 800 hours and a sample standard deviation of 70 hours. (i) Construct a 95% two-sided confidence interval for the mean life of the sensors. (ii) Construct a 99% two-sided confidence interval for the variance of the life of the sensors. (iii) If the population standard deviation is known to be 90 hours, construct a 99% two-sided confidence interval for the mean life. (iv) Comment on the answers to (ii) and (iii).	10	L4	CO3

(i) σ unknown, $t_{0.025,29} = 2.0452$: $800 \pm 2.0452\frac{70}{\sqrt{30}} = 800 \pm 26.14$, i.e. $773.86 \leq \mu \leq 826.14$ h.

(ii) $(n-1)s^2 = 29(4900) = 142100$, $\chi^2_{0.005,29} = 52.336$, $\chi^2_{0.995,29} = 13.121$:

$$\frac{142100}{52.336} \leq \sigma^2 \leq \frac{142100}{13.121} \Rightarrow 2715.2 \leq \sigma^2 \leq 10829.8$$

(so $52.11 \leq \sigma \leq 104.07$ h).

(iii) $\sigma = 90$, $z_{0.005} = 2.5758$: $800 \pm 2.5758\frac{90}{\sqrt{30}} = 800 \pm 42.33$, i.e. $757.67 \le \mu \le 842.33$ h.

(iv) The stated $\sigma^2 = 8100$ lies inside the 99% interval for σ^2 in (ii), so the sample is consistent with $\sigma = 90$ h. The interval in (iii) is wider than that in (i) because it uses 99% confidence and the larger known $\sigma = 90$ instead of $s = 70$. The variance interval is wide and not symmetric about $s^2 = 4900$ because the χ^2 distribution is skewed.

5	For the sample (6, 3, 2, 1, 1, 0, 1, 3, 3, 2) of the number of defects in appliances returned by customers, (i) construct a 90% confidence interval on the population mean number of defects per appliance, if the population standard deviation is known to be 2 defects; (ii) what should the sample size be to keep the width of the confidence interval on the mean below 2? (iii) construct a 95% confidence interval on the population mean, assuming the population variance is unknown.	10	L3	CO3

$n = 10$, $\bar{x} = 2.2$, $s^2 = 2.8444$, $s = 1.6865$.

(i) $z_{0.05} = 1.6449$: $2.2 \pm 1.6449\frac{2}{\sqrt{10}} = 2.2 \pm 1.04$, i.e. $1.16 \le \mu \le 3.24$.

(ii) Width $= 2z_{0.05}\frac{\sigma}{\sqrt{n}} < 2 \Rightarrow \sqrt{n} > 1.6449(2) = 3.29 \Rightarrow n > 10.82$, so $n = 11$.

(iii) $t_{0.025,9} = 2.2622$: $2.2 \pm 2.2622\frac{1.6865}{\sqrt{10}} = 2.2 \pm 1.206$, i.e. $0.994 \le \mu \le 3.406$.

6	The numbers of failed specimens per day in a load test, collected on 10 randomly chosen days, are: 100, 115, 150, 170, 90, 135, 180, 110, 150, 155. Assuming the numbers of failed specimens are approximately normal, test whether the mean number of failed specimens is at most 120, at the 10% significance level.	10	L4	CO4

$H_0 : \mu = 120$ (at most 120) against $H_1 : \mu > 120$; $\alpha = 0.10$; σ unknown, so use the t-test. $\bar{x} = 135.5$, $s = 30.5$.

$$t_0 = \frac{\bar{x} - 120}{s/\sqrt{n}} = \frac{15.5}{30.5/\sqrt{10}} = 1.607.$$

Critical value $t_{0.10,9} = 1.383$. Since $1.607 > 1.383$ (one-sided P-value $= 0.0713 < 0.10$), H_0 is rejected: at the 10% level the mean number of failed specimens exceeds 120.

7	The number of defects in printed circuit boards is hypothesised to follow a Poisson distribution. A random sample of $n = 60$ boards gives the following observed frequencies. Perform a test of the hypothesis. No. of defects: 0, 1, 2, 3 Observed frequency: 32, 15, 9, 4	10	L4	CO4

H_0: the number of defects is Poisson; H_1: it is not; $\alpha = 0.05$. The mean is estimated from the data: $\hat{\lambda} = \frac{0(32)+1(15)+2(9)+3(4)}{60} = \frac{45}{60} = 0.75$.

Defects x	**0**	**1**	**2**	≥ 3
$p(x)$	0.4724	0.3543	0.1329	0.0405
Expected $60p(x)$	28.34	21.26	7.97	2.43
Observed	32	15	9	4

The expected count in the last cell is below 3, so the last two cells are combined: expected 10.4, observed 13.

$$\chi_0^2 = \frac{(32 - 28.34)^2}{28.34} + \frac{(15 - 21.26)^2}{21.26} + \frac{(13 - 10.4)^2}{10.4} = 2.963.$$

Degrees of freedom $= k - p - 1 = 3 - 1 - 1 = 1$ (one parameter estimated). $\chi^2_{0.05,1} = 3.84$. Since $2.96 < 3.84$ (P-value $= 0.085$), H_0 is not rejected: the Poisson distribution is an adequate model for the defects.

8	Test at the 10% significance level whether the variance of a population is greater than 10, using the sample data 24, 22, 25, 23, 20, 23, 21, 22, 25, 22 (assume a normal population).	10	L4	CO4

$H_0 : \sigma^2 = 10$ against $H_1 : \sigma^2 > 10$; $\alpha = 0.10$. $\bar{x} = 22.7$, $\sum(x - \bar{x})^2 = 24.1$, $s^2 = 2.6778$.

$$\chi_0^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{24.1}{10} = 2.41.$$

Reject H_0 if $\chi_0^2 > \chi_{0.10,9}^2 = 14.684$. Since $2.41 < 14.684$ (P-value $= 0.983$), H_0 is not rejected: there is no evidence that the variance exceeds 10 (the sample variance is only 2.68).

9	A 1992 article in the *Journal of the American Medical Association* reported body temperatures for a number of subjects. The body temperatures (°F) of 25 female subjects were: 97.8, 97.2, 97.4, 97.6, 97.8, 97.9, 98.0, 98.0, 98.0, 98.1, 98.2, 98.3, 98.3, 98.4, 98.4, 98.4, 98.5, 98.6, 98.6, 98.7, 98.8, 98.8, 98.9, 98.9, 99.0. Test $H_0 : \mu = 98.6$ against $H_1 : \mu \neq 98.6$ using $\alpha = 0.05$.	10	L4	CO4

σ unknown; assume normal temperatures and use the t-test with $n - 1 = 24$ degrees of freedom. $\bar{x} = 98.264$, $s = 0.4821$.

$$t_0 = \frac{\bar{x} - 98.6}{s/\sqrt{25}} = \frac{-0.336}{0.4821/5} = -3.485.$$

Reject H_0 if $|t_0| > t_{0.025,24} = 2.064$. Since $|t_0| = 3.485 > 2.064$ (P-value $= 0.0019$), H_0 is rejected: the mean body temperature of women differs from 98.6°F (it is lower).

10	A student collected the sample data 24, 22, 25, 23, 20, 23, 21, 22, 25, 22 during an experiment. (i) What are the estimates of the population mean and population variance, and of the variance of the sample mean for samples of size 10? (ii) What is a 90% lower bound for the population mean? (iii) What is a 95% upper bound for the population variance?	10	L3	CO3

(i) $\hat{\mu} = \bar{x} = 22.7$; $\hat{\sigma}^2 = s^2 = \frac{24.1}{9} = 2.6778$; $\widehat{V}(\bar{X}) = \frac{s^2}{n} = 0.2678$.
(ii) $t_{0.10,9} = 1.383$: $\mu \geq \bar{x} - t_{0.10,9}\frac{s}{\sqrt{n}} = 22.7 - 1.383\frac{1.6364}{\sqrt{10}} = 21.984$.
(iii) $\chi_{0.95,9}^2 = 3.325$: $\sigma^2 \leq \frac{(n-1)s^2}{\chi_{0.95,9}^2} = \frac{24.1}{3.325} = 7.248$.

11	An engineer studying the tensile strength of a steel alloy intended for golf club shafts knows that tensile strength is approximately normally distributed. A random sample of 12 specimens has a mean tensile strength of 3250 psi. (a) Test the hypothesis that the mean strength is 3500 psi, at the 1% significance level; the standard deviation is known to be 60 psi. (b) What is the smallest level of significance at which you would reject the null hypothesis in part (a)? (c) Explain how you could answer part (a) with a two-sided confidence interval on the mean tensile strength. (d) How would the test be performed if the standard deviation were not known?	10	L4	CO4

(a) $H_0 : \mu = 3500$, $H_1 : \mu \neq 3500$, $\alpha = 0.01$. $z_0 = \dfrac{3250 - 3500}{60/\sqrt{12}} = \dfrac{-250}{17.321} = -14.43$. Since $|z_0| > z_{0.005} = 2.576$, H_0 is rejected.
(b) The P-value $= 2\Phi(-14.43) \approx 3.2 \times 10^{-47}$, practically zero, so H_0 would be rejected at any reasonable significance level.
(c) The 99% interval is $3250 \pm 2.576(17.321) = 3250 \pm 44.61$, i.e. $3205.39 \leq \mu \leq 3294.61$. Since 3500 lies outside the

interval, H_0 is rejected at $\alpha = 0.01$, the same conclusion.

(d) Use the sample standard deviation s and the t-test, $t_0 = \frac{\bar{x}-3500}{s/\sqrt{12}}$, compared with $t_{0.005,11} = 3.106$.

12	The sugar content of the syrup in canned peaches is normally distributed. A random sample of $n = 10$ cans gives a sample standard deviation of $s = 4.8$ mg. (i) Find a 90% two-sided confidence interval for the population standard deviation. (ii) Find a 95% upper bound and a 95% lower bound for the population standard deviation. How do these bounds compare with the limits in part (i)?	10	L3	CO3

$(n-1)s^2 = 9(23.04) = 207.36$; $\chi^2_{0.05,9} = 16.919$, $\chi^2_{0.95,9} = 3.325$.

(i) $\frac{207.36}{16.919} \le \sigma^2 \le \frac{207.36}{3.325} \Rightarrow 12.256 \le \sigma^2 \le 62.362$, so $3.501 \le \sigma \le 7.897$ mg.

(ii) 95% upper bound: $\sigma \le \sqrt{207.36/3.325} = 7.897$ mg; 95% lower bound: $\sigma \ge \sqrt{207.36/16.919} = 3.501$ mg. They are exactly the limits of the 90% two-sided interval, because each one-sided 95% bound puts 5% in one tail, as each end of the 90% interval does.

13	Cloud seeding has been studied for many decades as a weather-modification method. The rainfall (acre-feet) from 20 clouds selected at random and seeded with silver nitrate was: 18.0, 30.7, 19.8, 27.1, 22.3, 18.8, 31.8, 23.4, 21.2, 27.9, 31.9, 27.1, 25.0, 24.7, 26.9, 21.8, 29.2, 34.8, 26.7, 31.6. (a) Construct a 95% one-sided lower confidence bound on the mean rainfall. (b) Construct a 90% two-sided confidence interval on the mean rainfall. What is the width of this interval? (c) If the width of the interval in part (b) is to be halved, how large should the sample be? Use trial and error if needed.	10	L4	CO3

Assume normal rainfall. $n = 20$, $\bar{x} = 26.035$, $s = 4.7848$; $t_{0.05,19} = 1.7291$, so $t_{0.05,19}\, s/\sqrt{20} = 1.85$.

(a) $\mu \ge 26.035 - 1.85 = 24.185$ acre-feet.

(b) 26.035 ± 1.85, i.e. $24.185 \le \mu \le 27.885$; width $= 3.7$ acre-feet.

(c) The width varies as $t/\sqrt{n}$, so halving it needs roughly $4 \times 20 = 80$ observations. With s unchanged, the required half-width is 0.925: $n \ge \left(t\, s/0.925\right)^2$; trial with $t_{0.05,n-1}$ gives $n = 75$ (e.g. $t_{0.05,74} = 1.6657$ and $1.6657(4.785)/\sqrt{75} = 0.92$).

14	Amino acid concentrations in 8 packs of dog feed were recorded in mg/kg of feed as: 22, 24, 20, 26, 27, 24, 26, 23. (i) Perform a test at the 5% significance level to see whether the population variance is greater than 5.5 (mg/kg)^2. What is the P-value of this test? (ii) Construct 90% upper and 90% lower confidence bounds for the population variance.	10	L4	CO4

Assume a normal population. $\bar{x} = 24$, $\sum(x-\bar{x})^2 = 38$, $s^2 = 5.4286$.

(i) $H_0 : \sigma^2 = 5.5$, $H_1 : \sigma^2 > 5.5$. $\chi^2_0 = \frac{38}{5.5} = 6.909$; $\chi^2_{0.05,7} = 14.067$. Since $6.909 < 14.067$, H_0 is not rejected. P-value $= P(\chi^2_7 > 6.909) = 0.438$.

(ii) With $\alpha = 0.10$ in one tail: upper bound $\sigma^2 \le \frac{38}{\chi^2_{0.90,7}} = \frac{38}{2.833} = 13.413$; lower bound $\sigma^2 \ge \frac{38}{\chi^2_{0.10,7}} = \frac{38}{12.017} = 3.162$ (mg/kg)^2.

15	Define and interpret a confidence interval with examples.	06	L2	CO3

A $100(1-\alpha)\%$ confidence interval for a parameter θ is an interval $l \leq \theta \leq u$ computed from sample data by a procedure that captures the true θ in $100(1-\alpha)\%$ of repeated samples. The parameter is fixed; it is the interval that varies from sample to sample. So a 95% interval does not mean that θ lies in this particular interval with probability 0.95, but that the method succeeds 95% of the time. Its width shows the precision of the estimate: it narrows as n increases and widens as the confidence level increases.
Examples: (1) a sample of 36 has $\bar{x} = 52$ with $\sigma = 6$: the 95% interval $50.04 \leq \mu \leq 53.96$ means we are 95% confident that the mean lies between 50.04 and 53.96. (2) For 30 sensors with $\bar{x} = 800$ h and $s = 70$ h, the 95% t-interval 773.9 h to 826.1 h estimates the mean sensor life.

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